



DEPARTMENT OF PUBLIC HEALTH AND ENVIRONMENT

Air Quality Control Commission

REGULATION NUMBER 26

CONTROL OF EMISSIONS FROM ENGINES AND MAJOR STATIONARY SOURCES

5 CCR 1001-30

[Editor's Notes follow the text of the rules at the end of this CCR Document.]

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Pursuant to Colorado Revised Statutes § 24-4-103 (12.5), materials incorporated by reference are available for public inspection during normal business hours, or copies may be obtained at a reasonable cost from the Air Quality Control Commission (the Commission), 4300 Cherry Creek Drive South, Denver, Colorado 80246-1530. The material incorporated by reference is also available through the United States Government Printing Office, online at www.govinfo.gov. Materials incorporated by reference are those editions in existence as of the date indicated and do not include any later amendments.

PART A General Provisions

I. General Provisions

I.A. Definitions

I.A.1. “8-Hour Ozone Control Area” means the Counties of Adams, Arapahoe, Boulder (includes part of Rocky Mountain National Park), Douglas, and Jefferson; the Cities and Counties of Denver and Broomfield; and the following portions of the Counties of Larimer and Weld:

I.A.1.a. For Larimer County (includes part of Rocky Mountain National Park), that portion of the county that lies south of a line described as follows: Beginning at a point on Larimer County’s eastern boundary and Weld County’s western boundary intersected by 40 degrees, 42 minutes, and 47.1 seconds north latitude, proceed west to a point defined by the intersection of 40 degrees, 42 minutes, 47.1 seconds north latitude and 105 degrees, 29 minutes, and 40.0 seconds west longitude, thence proceed south on 105 degrees, 29 minutes, 40.0 seconds west longitude to the intersection with 40 degrees, 33 minutes and 17.4 seconds north latitude, thence proceed west on 40 degrees, 33 minutes, 17.4 seconds north latitude until this line intersects Larimer County’s western boundary and Grand County’s eastern boundary.

I.A.1.b. For Weld County, that portion of the county that lies south of a line described as follows: Beginning at a point on Weld County’s eastern boundary and Logan County’s western boundary intersected by 40 degrees, 42 minutes, 47.1 seconds north latitude, proceed west on 40 degrees, 42 minutes, 47.1 seconds north latitude until this line intersects Weld County’s western boundary and Larimer County’s eastern boundary.

- I.A.2. "Denver 1-Hour Ozone Attainment/Maintenance Area" means the Counties of Jefferson and Douglas, the Cities and Counties of Denver and Broomfield, Boulder County (excluding Rocky Mountain National Park), Adams County west of Kiowa Creek, and Arapahoe County west of Kiowa Creek.
- I.A.3. "Northern Weld County" means the portion of the county that does not lie south of a line described as follows: Beginning at a point on Weld County's eastern boundary and Logan County's western boundary intersected by 40 degrees, 42 minutes, 47.1 seconds north latitude, proceed west on 40 degrees, 42 minutes, 47.1 seconds north latitude until this line intersects Weld County's western boundary and Larimer County's eastern boundary.
- I.A.4. "Volatile Organic Compound (VOC)" means any compound of carbon, excluding carbon monoxide, carbon dioxide, carbonic acid, metallic carbides or carbonates, and ammonium carbonate, which participates in atmospheric photochemical reactions, except those listed in Section II.B. as having negligible photochemical reactivity. VOC may be measured by a reference method, an equivalent method, an alternative method, or by procedures specified under 40 CFR Part 60 (July 1, 2022). A reference method, an equivalent method, or an alternative method, however, may also measure nonreactive organic compounds. In such cases, an owner or operator may exclude the compounds listed in Section II.B. when determining compliance with a standard if the amount of such compounds is accurately quantified, and such exclusion is approved by the Division. As a precondition to excluding such compounds as VOC, or at any time thereafter, the Division may require an owner or operator to provide monitoring or testing methods and results demonstrating, to the satisfaction of the Division, the amount of negligible-reactive compounds in the source's emissions.

I.B. Exemptions

Emissions of the organic compounds listed as having negligible photochemical reactivity in the common provisions definition of Negligibly Reactive Volatile Organic Compound are exempt from the provisions of this regulation.

I.C. New Sources

All new sources shall utilize controls representing RACT, pursuant to applicable provisions in Regulation Number 7, Regulation Number 24, Regulation Number 25, Regulation Number 26 and Regulation Number 3, Part B, Section III.D., upon commencement of operation.

I.D. Alternative Control Plans and Test Methods

I.D.1. Sources subject to specific requirements of this regulation shall submit for approval as a revision to the State Implementation Plan:

I.D.1.a. Any alternative emission control plan or compliance method other than control options specifically allowed in the applicable regulation. Such alternative control plans shall provide control equal to or greater than the emission control or reduction required by the regulation, unless the source contends that the control level required by the regulation does not represent RACT for their specific source.

I.D.1.b. Any alternative test method or procedure not specifically allowed in the applicable regulation.

I.D.2. No alternative submitted pursuant to this Section II.D. is effective until the alternative is approved as a revision to the State Implementation Plan.

I.E. The provisions marked by (State Only) are not federally enforceable, unless otherwise identified.

Appendix A Colorado Ozone Nonattainment or Attainment Maintenance Areas

I. Chronology of Attainment Status

Denver Metropolitan Area Only

1978 Denver 1-hour Ozone Nonattainment Area designation first becomes effective in 7-county Denver Metropolitan Area

10/11/01 Denver 1-hour Ozone Attainment Maintenance Area designation replaces non-attainment designation and becomes effective in 7-county Denver Metropolitan Area

9/2/05 1-hour Ozone National Ambient Air Quality Standard is Revoked in Colorado except for the Denver 1-hour Ozone Attainment Maintenance Area.

Denver Metropolitan Area and North Front Range

10/11/01 1-hour attainment maintenance area replaces non-attainment designation for the Denver Metro Area/North Front Range Area

4/15/04 EPA designates the Denver Metro Area/North Front Range region as an 8-hour ozone non-attainment area, designation deferred due to the implementation of the Early Action Compact

11/20/07 Denver 8-hour ozone non-attainment designation (1997 NAAQS) becomes effective in 9 county Denver Metropolitan Area

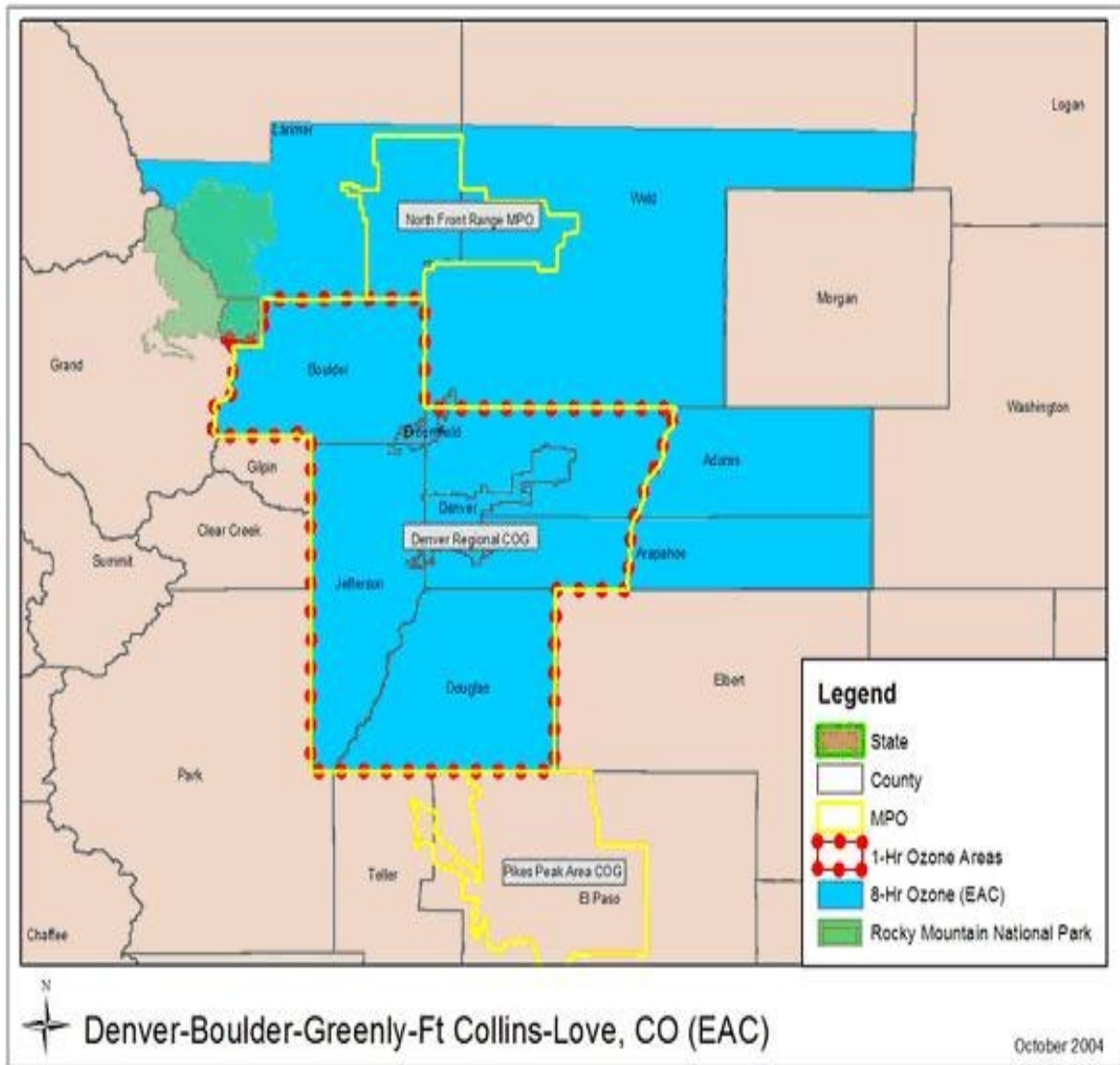
7/20/2012 Denver 8-hour ozone non-attainment designation (2008 NAAQS) becomes effective in 9 county Denver Metropolitan Area

8/3/2018 Denver 8-hour ozone nonattainment designation (2015 NAAQS) becomes effective in 9 county Denver Metropolitan Area

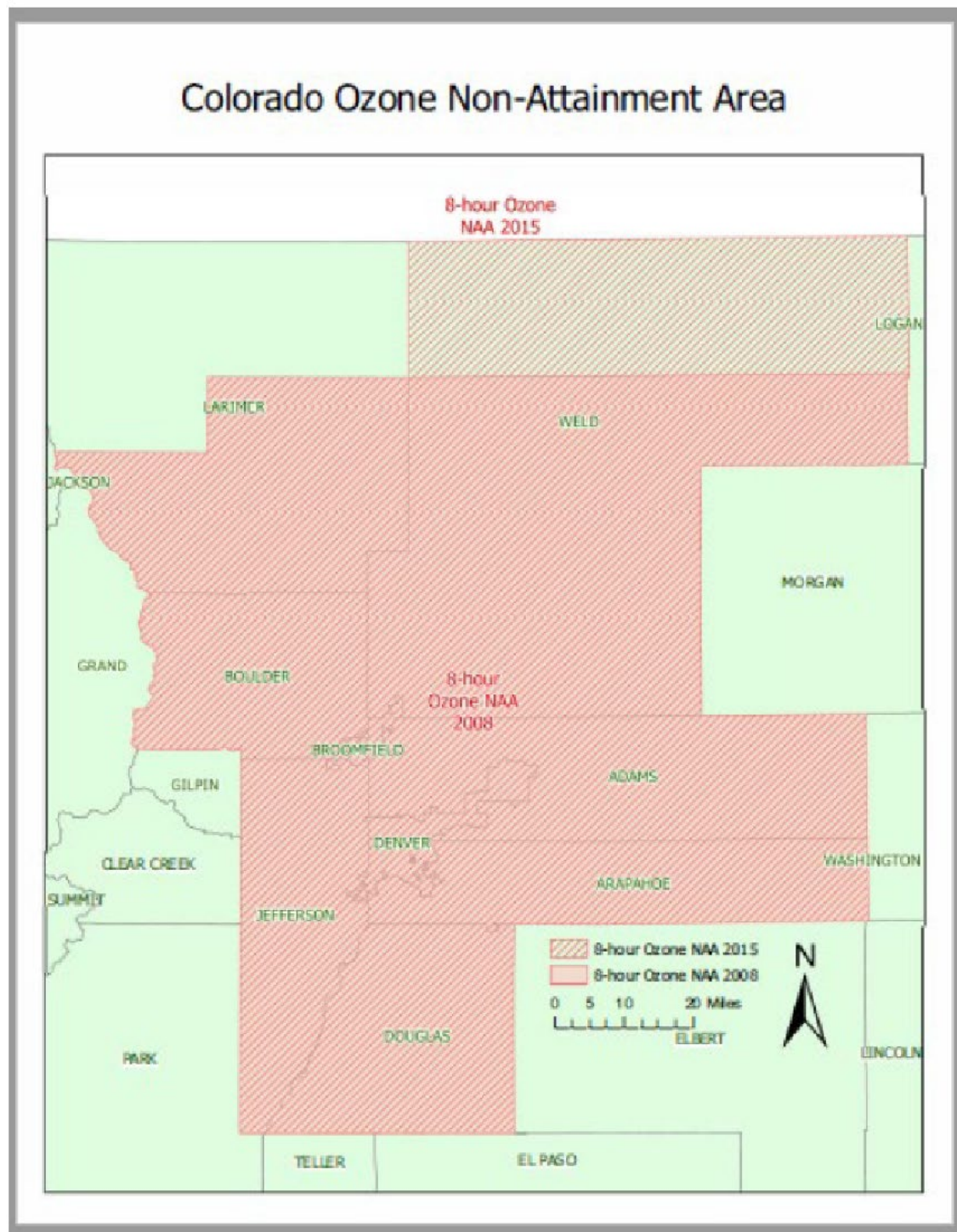
12/31/2021 EPA modification of the 9 county Denver Metropolitan Area 8-hour ozone nonattainment designation (2015 NAAQS) to include the portion of northern Weld County defined in Part A

II. Maps

Denver Metropolitan Area and North Front Range (2008 Ozone NAAQS)



Denver Metropolitan Area and North Front Range and northern Weld County (2015 ozone NAAQS)



PART B Combustion Equipment and Major Source RACT

I. Control of Emissions from Engines

I.A. Requirements for new and existing engines.

- I.A.1. The owner or operator of any natural gas-fired stationary or portable reciprocating internal combustion engine with a manufacturer's design rate greater than 500 horsepower commencing operations in the 8-hour Ozone Control Area on or after June 1, 2004 shall employ air pollution control technology to control emissions, as provided in Section I.B.
- I.A.2. Any existing natural gas-fired stationary or portable reciprocating internal combustion engine with a manufacturer's design rate greater than 500 horsepower, which existing engine was operating in the 8-hour Ozone Control Area prior to June 1, 2004, shall employ air pollution control technology on and after May 1, 2005, as provided in Section I.B.
- I.A.3. Beginning January 14, 2026, the requirements in Section I.B. are also SIP applicable in northern Weld County for lean burn reciprocating internal combustion engines with a manufacturer's nameplate design rate greater than 500 horsepower that were constructed or modified before February 1, 2009, which were required to install and operate an oxidation catalyst by July 1, 2010, pursuant to Section I.D.4.b.
- I.A.4. Stationary natural gas fired reciprocating internal combustion engines state-wide with a manufacturer's design rate greater than or equal to 1000 horsepower are subject to Section I.D.5.

I.B. Air pollution control technology requirements

- I.B.1. For rich burn reciprocating internal combustion engines, a non-selective catalyst reduction and an air fuel controller shall be required. A rich burn reciprocating internal combustion engine is one with a normal exhaust oxygen concentration of less than 2% by volume.
- I.B.2. For lean burn reciprocating internal combustion engines, an oxidation catalyst shall be required. A lean burn reciprocating internal combustion engine is one with a normal exhaust oxygen concentration of 2% by volume, or greater.
- I.B.3. The emission control equipment required by this Section I.B shall be appropriately sized for the engine and shall be operated and maintained according to manufacturer specifications.

- I.C. The air pollution control technology requirements in Sections I.A. and I.B. do not apply to:
- I.C.1. Non-road engines, as defined in Regulation Number 3, Part A, Section I.B.36.
 - I.C.2. Reciprocating internal combustion engines that the Division has determined will be permanently removed from service or replaced by electric units on or before May 1, 2007. The owner or operator of such an engine shall provide notice to the Division of such intent by May 1, 2005 and shall not operate the engine identified for removal or replacement in the 8-hour Ozone Control Area after May 1, 2007.
 - I.C.3. Any emergency power generator exempt from APEN requirements pursuant to Regulation Number 3, Part A.
 - I.C.4. Any lean burn reciprocating internal combustion engine operating in the 8-hour Ozone Control Area prior to June 1, 2004, for which the owner or operator demonstrates to the Division that retrofit technology cannot be installed at a cost of less than \$5,000 per ton of VOC emission reduction. Installation costs and the best information available for determining control efficiency shall be considered in determining such costs. In order to qualify for such exemption, the owner or operator must submit an application making such a demonstration, together with all supporting documents, to the Division by May 1, 2005. Any reciprocating internal combustion engine qualifying for this exemption shall not be moved to any other location within the 8-hour Ozone Control Area.
- I.D. Control of emissions from new, modified, existing, and relocated natural gas fired reciprocating internal combustion engines.
- I.D.1. (State Only) Exemptions
 - I.D.1.a. The requirements of this Section I.D. do not apply to any engine having actual uncontrolled emissions below permitting thresholds listed in Regulation Number 3, Part B.
 - I.D.1.b. Internal combustion engines that are subject to an emissions control requirement in a federally maximum achievable control technology (MACT) standard under 40 CFR Part 63 (July 1, 2022), a Best Available Control Technology (BACT) limit, or a New Source Performance Standard (NSPS) under 40 CFR Part 60 (July 1, 2022) are not subject to Section I.D.3.
 - I.D.2. (State Only) General Provisions

I.D.2.a. At all times, including periods of start-up and shutdown, engines and their associated equipment must be maintained and operated in a manner consistent with good air pollution control practices for minimizing emissions. Determination of whether or not acceptable operation and maintenance procedures are being used will be based on information available to the Division, which may include, but is not limited to, monitoring results, opacity observations, review of operation and maintenance procedures, and inspection of the source.

I.D.2.b. All engines and their associated equipment must be operated and maintained pursuant to the manufacturing specifications or equivalent to the extent practicable, and consistent with technological limitations and good engineering and maintenance practices. The owner or operator must keep manufacturer specifications or equivalent on file.

I.D.2.c. Any of the effective dates for installation of controls on internal combustion engines as required in Section I.D.3. may be extended at the Division's discretion for good cause shown.

I.D.3. (State Only) New, Modified and Relocated Natural Gas Fired Reciprocating Internal Combustion Engines

I.D.3.a. Except as provided in Section I.D.3.b., the owner or operator of any natural gas fired reciprocating internal combustion engine that is either constructed or relocated to the state of Colorado from another state, on or after the date listed in Table 1 shall operate and maintain each engine according to the manufacturer's written instructions or procedures to the extent practicable and consistent with technological limitations and good engineering and maintenance practices over the entire life of the engine so that it achieves the emission standards required in Section I.D.3.b. Table 1.

I.D.3.b. Actual emissions from natural gas fired reciprocating internal combustion engines shall not exceed the emission performance standards in Table 1 as expressed in units of grams per horsepower-hour (G/hp-hr)

TABLE 1				
Maximum Engine Hp	Construction or Relocation Date	Emission Standards is G/hp-hr		
		NOx	CO	VOC
< 100 Hp	Any	NA	NA	NA
≥ 100 Hp and < 500 Hp	On or after January 1, 2008	2.0	4.0	1.0
	On or after January 1, 2011	1.0	2.0	0.7
≥ 500 Hp	On or after July 1, 2007	2.0	4.0	1.0
	On or after July 1, 2010	1.0	2.0	0.7

**These engines may also be subject to emission standards under Section I.D.5.*

I.D.4. Existing Natural Gas Fired Reciprocating Internal Combustion Engines

I.D.4.a. (Regional Haze SIP) Rich Burn Reciprocating Internal Combustion Engines

I.D.4.a.(i) Except as provided in Sections I.D.4.a.(i)(B) and (C) and I.E.4.a.(ii), all rich burn reciprocating internal combustion engines with a manufacturer's name plate design rate greater than 500 horsepower, constructed or modified before February 1, 2009 shall install and operate both a non-selective catalytic reduction system and an air fuel controller by July 1, 2010. A rich burn reciprocating internal combustion engine is one with a normal exhaust oxygen concentration of less than 2% by volume.

I.D.4.a.(i)(A) All control equipment required by this Section I.D.4.a. shall be operated and maintained pursuant to manufacturer specifications or equivalent to the extent practicable, and consistent with technological limitations and good engineering and maintenance practices. The owner or operator shall keep manufacturer specifications or equivalent on file.

I.D.4.a.(i)(B) Internal combustion engines that are subject to an emissions control requirement in a federal maximum achievable control technology ("MACT") standard under 40 CFR Part 63 (July 1, 2022), a Best Available Control Technology ("BACT") limit, or a New Source Performance Standard under 40 CFR Part 60 (July 1, 2022) are not subject to this Section I.D.4.a.

I.D.4.a.(i)(C) The requirements of this Section I.D.4.a. do not apply to any engine having actual uncontrolled emissions below permitting thresholds listed in Regulation Number 3, Part B.

I.D.4.a.(ii) Any rich burn reciprocating internal combustion engine constructed or modified before February 1, 2009, for which the owner or operator demonstrates to the Division that retrofit technology cannot be installed at a cost of less than \$ 5,000 per ton of combined volatile organic compound and nitrogen oxides emission reductions (this value shall be adjusted for future applications according to the current day consumer price index) is exempt complying with Section I.D.4.a. Installation costs and the best information available for determining control efficiency shall be considered in determining such costs. In order to qualify for such exemption, the owner or operator must submit an application making such a demonstration, together with all supporting documents, to the Division by August 1, 2009.

I.D.4.b. (State Only) Lean Burn Reciprocating Internal Combustion Engines

I.D.4.b.(i) Except as provided in Section I.D.4.b.(ii), all lean burn reciprocating internal combustion engines with a manufacturer's nameplate design rate greater than 500 horsepower shall install and operate an oxidation catalyst by July 1, 2010. A lean burn reciprocating internal combustion engine is one with a normal exhaust oxygen concentration of 2% by volume, or greater.

I.D.4.b.(ii) Any lean burn reciprocating internal combustion engine constructed or modified before February 1, 2009, for which the owner or operator demonstrates to the Division that retrofit technology cannot be installed at a cost of less than \$ 5,000 per ton of volatile organic compound emission reduction (this value shall be adjusted for future applications according to the current day consumer price index) is exempt complying with Section I.D.4.b.(i). Installation costs and the best information available for determining control efficiency shall be considered in determining such costs. In order to qualify for such exemption, the owner or operator must submit an application making such a demonstration, together with all supporting documents, to the Division by August 1, 2009.

I.D.4.c. (Ozone SIP) Engines with a Manufacturer's Design Rate Greater than or Equal to 1000 horsepower in the 8-hour Ozone Control Area or Northern Weld County.

I.D.4.c.(i) The owner or operator of an engine identified in Table A must comply with the current NO_x allowance specified for that engine by the compliance date indicated.

I.D.4.c.(ii) Owners or operators of engines identified in Table A must keep the following records for a period of five (5) years and make records available to the Division upon request.

I.D.4.c.(ii)(A) APENs submitted for a Table A engine on or after May 1, 2021.

I.D.4.c.(ii)(B) Records documenting engine retrofit or replacement.

I.D.4.c.(ii)(C) Results of the most recent performance test if updated permitted emission factors are used to comply with the NO_x emission requirements.

Table A					
Facility AIRS ID	AIRs Point(s)	Prior NOx allowanc e (tpy)_	Current NOx allowanc e (tpy)	Emission Reduction method	Compliance Date
001-0025	004	234.2	32.2	Retrofit	05/2023
001-0025	006	128.0	15.9	Retrofit	05/2023
001-0025	007	128.0	15.9	Retrofit	05/2022
01-0025	032	234.2	32.2	Retrofit	05/2022
001-0036	005	150.50	8.6	Permitted Hours Reduced	05/2023
001-0036	005	150.50	8.6	Permitted Hours Reduced	05/2023
005-0055	001	148.92	20.6	Retrofit	05/2022
005-0055	001	148.92	20.6	Retrofit	05/2022
005-0055	001	148.92	20.6	Retrofit	05/2022
005-0055	011	133.78	18.6	Retrofit	05/2022
123-0015	051	11.9	9.5	Retrofit	05/2022
123-0015	052	24.0	9.5	Retrofit	05/2022
123-0015	053	23.8	9.5	Retrofit	05/2022
123-0015	055	23.8	9.5	Retrofit	05/2022
123-0015	056	11.9	9.5	Retrofit	05/2022
123-0015	057	24.0	9.5	Retrofit	05/2022
123-0015	060	19.3	7.7	Retrofit	05/2022

Table A					
Facility AIRS ID	AIRs Point(s)	Prior NOx allowanc e (tpy)_	Current NOx allowanc e (tpy)	Emission Reduction method	Compliance Date
123-0015	081	23.8	9.5	Retrofit	05/2022
123-0048	015	16.9	6.3	Retrofit	05/2022
123-0049	101	21.2	8.5	Retrofit	05/2023
123-0049	102	7.8	6.2	Retrofit	05/2023
123-0049	103	26.1	10.4	Retrofit	05/2023
123-0049	107	16.3	6.2	Retrofit	05/2023
123-0049	108	13.0	10.4	Retrofit	05/2023
123-0049	110	18.6	7.1	Retrofit	05/2023
123-0049	113	16.3	6.2	Retrofit	05/2023
123-0049	114	7.8	6.2	Retrofit	05/2023
123-0049	140	14.3	11.4	Retrofit	05/2023
123-0098	001	18.19	10.51	Permitted EF Change	05/2021
123-0098	039	15.78	12.62	Permitted EF Change	05/2021
123-0099	106	21.24	8.5	Retrofit	05/2022
123-0099	107	24.0	8.5	Retrofit	05/2022
123-0099	108	8.65	6.9	Retrofit	05/2022
123-0099	109	21.24	8.5	Retrofit	05/2022
123-0099	110	24.0	8.5	Retrofit	05/2022

Table A					
Facility AIRS ID	AIRs Point(s)	Prior NOx allowance (tpy)	Current NOx allowance (tpy)	Emission Reduction method	Compliance Date
123-0184	018	25.1	8.7	Retrofit	05/2022
123-0184	019	8.7	6.9	Retrofit	05/2022
123-0468	011	18.9	15.5	Retrofit	05/2022
123-0468	012	19.5	15.5	Replace	05/2022
123-0595	002	30.82	13.0	Retrofit	05/2023
123-0595	003	30.82	13.0	Retrofit	05/2023
123-0595	004	30.82	13.0	Retrofit	05/2023
123-0595	006	25.69	10.8	Retrofit	05/2023
123-0595	007	25.69	10.8	Retrofit	05/2023
123-0595	008	27.12	11.4	Retrofit	05/2023
123-1351	005	12.8	10.4	Retrofit	05/2022
123-9E80	008	0.00	0.00	Replace	05/2022

I.D.4.c.(iii) The owner or operator of an engine identified in Table B must shutdown that engine by the compliance date indicated.

I.D.4.c.(iv) Owners or operators of engines identified in Table B must keep the following records for a period of five (5) years and make records available to the Division upon request.

I.D.4.c.(iv)(A) APENs cancellations submitted for a Table B engine on or after May 1, 2021.

Table B			
Facility AIRS ID	AIRs Point(s)	Previously Permitted NOx allowance (tpy)	Shutdown Compliance Date
123-0057	004	90.2	05/2022
123-1351	001	15.8	05/2022

I.D.5. (State Only) Additional Requirements for Natural Gas Fired Reciprocating Internal Combustion Engines

I.D.5.a. Applicability

I.D.5.a.(i) This Section I.D.5. applies to stationary natural gas fired reciprocating internal combustion engines state-wide with a manufacturer's design rate greater than or equal to 1000 horsepower.

I.D.5.a.(i)(A) For purposes of this Section I.D.5., modified means any physical change to the engine or change in method of operation that results in an increase in the emission rate of any air pollutant, and does not include any physical or operational changes excluded by 40 C.F.R. 60.14(e).

I.D.5.a.(i)(B) For purposes of this Section I.D.5., placed in service means the bringing of an engine on-site for use. The placed in service date is the date the engine begins to operate. The following is not considered placed in service: (1) moving an engine subject to an Alternative Company-Wide Compliance Plan to another site with the same owner or operator; (2) for engines in service on or before January 30, 2024, replacement under an authorized alternative operating scenario.

I.D.5.a.(i)(C) For purposes of this Section I.D.5., relocated means the bringing of an engine into the 8-Hour Ozone Control Area from outside the 8-Hour Ozone Control Area or the bringing of an engine into the State of Colorado from outside the State of Colorado. The relocation date is the date the subject engine begins to operate.

I.D.5.a.(ii) Exemptions.

I.D.5.a.(ii)(A) Engines that burn less than 100 MMBtu per year of natural gas on a rolling-12-month basis are not subject to Sections I.D.5.b., I.D.5.d., I.D.5.e., I.D.5.f.(i)-(iii) and (v)-(vi), or I.D.5.g.

I.D.5.a.(ii)(B) Non-road engines, as defined in Regulation Number 3, Part A, Section I.B.36 are not subject to this Section I.D.5.

I.D.5.a.(ii)(C) Any emergency power generator exempt from APEN or construction permit requirements pursuant to Regulation Number 3, Parts A or B are not subject to this Section I.D.5.

I.D.5.a.(ii)(D) Emergency power generators that operate less than 250 hours per year on a rolling-12-month basis are not subject to Sections I.D.5.b., I.D.5.d., I.D.5.e., I.D.5.f.(i)-(iii) and (v)-(vi), or I.D.5.g.

I.D.5.b. Emission Standards for Engines Subject to Section I.D.5.a.

I.D.5.b.(i) The owner or operator of any stationary natural gas fired reciprocating internal combustion engine that is placed in service, modified, or relocated after November 14, 2020, must comply with the emission standards in Table 2 upon placement in service, modification, or relocation, as applicable.

I.D.5.b.(ii) The owner or operator of any stationary natural gas fired reciprocating internal combustion engine not subject to Section I.D.5.b.(i) must comply with the emission standards in Table 2 in accordance with the timing set forth Section I.D.5.b.(v).

TABLE 2			
Engine Type	Emission Standards (g/hp-hr)		
	NOx	CO	VOC
4-Stroke Lean Burn engines in service on or before November 14, 2020, (unless subject to a more stringent emission standard under Section I.D.3.b, above)	1.2	2.0	0.7
<u>Rich Burn engines in service on or before November 14, 2020</u>	0.8	2.0	0.7
<u>4-Stroke Lean Burn engines placed in service, modified, or relocated after November 14, 2020</u>	0.7	2.0	0.7
<u>Rich Burn engines placed in service, modified, or relocated after November 14, 2020</u>	0.5	2.0	0.7
<u>4-Stroke Lean Burn engines placed in service, modified, or relocated after January 30, 2024</u>	0.5	2.0	0.7
<u>2-Stroke Lean Burn engines</u>	3.0	2.0	0.7

I.D.5.b.(iii) By May 1, 2021, owners and operators of an engine placed in service on or before November 14, 2020, that is subject to an emission standard in Table 2 must submit a notification to the Division containing the following information:

- I.D.5.b.(iii)(A) The list of engines subject to an emission standard in Table 2, including AIRS number, location (inside or outside the 8-Hour Ozone Control Area and facility name), historical annual hours of operation averaged over calendar years 2017, 2018, and 2019, manufacturer model, serial number, horsepower, and engine configuration. The notification must also identify or calculate the g/hp-hr limit in an existing permit and the g/hp-hr at which the engine is operating on or before November 14, 2020, if different than the permitted rate. Engine configuration (e.g. rich burn or lean burn) for purposes of the emission standards in Table 2 is determined by the characterization on the engine's permit or APEN as of May 1, 2021. If the engine configuration is not identified in a permit or APEN, the owner or operator must submit an APEN with the current configuration information as determined by the owner or operator by May 1, 2021 to the Division.
- I.D.5.b.(iii)(B) An identification of the applicable standard and a declaration as to whether each subject engine meets the applicable standard as of May 1, 2021. If an engine will meet the applicable standard through a permit modification only, as described in Section I.D.5.b.(iv)(A), the declaration should note the date of permit modification submittal.
- I.D.5.b.(iii)(C) For all engines that do not meet the applicable emission standard as of May 1, 2021 or that cannot comply through a permit modification described in Section I.D.5.b.(iv)(A), a declaration of what action the owner or operator will take to meet the standard (e.g., control equipment installation, retrofit, replacement, electrification, shut-down). This declaration can be amended at any time prior to the applicable compliance date for that engine.

I.D.5.b.(iii)(D) The compliance deadline for each engine under Sections I.D.5.b.(i) or I.D.5.b.(v). An owner or operator may change a proposed compliance deadline for an engine subject to Section I.D.5.b.(v)(B) prior to that engine's compliance deadline, only after submittal of an updated notification to the Division that includes the updated compliance date and a demonstration that the requirements of Table 3 are met.

I.D.5.b.(iii)(E) Owners or operators that submit an Alternative Company-Wide Compliance Plan under Section I.D.5.c. are not subject to this Section I.D.5.b.(iii) for the emission standards in Table 2 for the engines covered by the Alternative Company-Wide Compliance Plan.

I.D.5.b.(iv) Permit Modification.

I.D.5.b.(iv)(A) An engine in service on or before November 14, 2020 that requires only a modification of an existing permit to meet the emission standards in this Section I.D.5.b. must submit a complete permit application containing the necessary limitations no later than May 1, 2021.

I.D.5.b.(iv)(B) For any engine not subject to Section I.D.5.b.(iv)(A), owners and operators must modify existing permits to reflect the emission standards or other operating conditions necessary to achieve compliance with Table 2. Complete permit applications must be submitted to the Division at least 365 days prior to the date established in Section I.D.5.b.(iii)(D) for that engine.

I.D.5.b.(v) Compliance Deadlines for engines subject to Section I.D.5.b.(ii).

I.D.5.b.(v)(A) Engines that comply with the emission standards on or before November 14, 2020, or are subject to Section I.D.5.b.(iv)(A) must meet the emission standards in Table 2 by May 1, 2022.

I.D.5.b.(v)(B) Engines not subject to Section I.D.5.b.(v)(A) must meet the emission standards in Table 2 in accordance with the timing set forth in Table 3.

TABLE 3					
Location of Subject Engines by Owner or Operator	Compliance Deadlines				
	May 1, 2022	May 1, 2023	May 1, 2024	May 1, 2025	May 1, 2026
	Percent (%) of engines that must comply with Table 2 limits				
Inside, or inside and outside, the 8-Hour Ozone Control Area	At least 34% of engines inside the 8-Hour Ozone Control Area	At least 67% of engines inside the 8-Hour Ozone Control Area; and at least 25% of engines outside the 8-Hour Ozone Control Area	100% of engines in the 8-Hour Ozone Control Area; and at least 50% of engines outside the 8-Hour Ozone Control Area	At least 75% of engines outside the 8-Hour Ozone Control Area	100% of all engines
Outside the 8-Hour Ozone Control Area only	At least 20%	At least 40%	At least 60%	At least 80%	100%

I.D.5.b.(vi) If an owner or operator replaces an engine subject to an emission standard under this Section I.D.5.b. with a different stationary natural gas fired reciprocating internal combustion engine, the replacement engine must:

I.D.5.b.(vi)(A) if being placed under an alternative operating scenario pursuant to an existing Division issued permit, meet the same emission standard as the engine being replaced; or

I.D.5.b.(vi)(B) if the owner or operator of an engine chooses to comply via an Alternative Company-Wide Compliance Plan under Section I.D.5.c., meet an emission standard at least as stringent as the engine being replaced as provided for in the applicable Alternative Company-Wide Compliance Plan.

I.D.5.c. Alternative Company-Wide Compliance Plan.

I.D.5.c.(i) Owners and operators with five or more engines that are subject to Section I.D.5.b.(v)(B) may comply with the NO_x requirements of Section I.D.5.b. through an Alternative Company-Wide Compliance Plan. Any owner or operator electing to develop an Alternative Company-Wide Compliance Plan must submit a Compliance Plan that meets the requirements of Section I.D.5.c.(ii) on or before May 1, 2021.

I.D.5.c.(i)(A) Only engines subject to an emission standard in Table 2 and that were placed in service on or before the November 14, 2020, can be included in an Alternative Company-Wide Compliance Plan submitted pursuant to this Section I.D.5.c.

I.D.5.c.(i)(B) Engines in an Alternative Company-Wide Compliance Plan must still meet the VOC and CO standards in Table 2 by the deadline established for that engine pursuant to Table 4.

I.D.5.c.(i)(C) Owners and operators owned by the same parent company may collectively submit a Compliance Plan in accordance with this Section I.D.5.c. However, the Compliance Plan must be signed and certified by a responsible official from each owner or operator with engines subject to the Compliance Plan acknowledging that each owner and operator is jointly and severally liable for compliance with the Compliance Plan and the provisions of this Section I.D.5.c. No engine may be included in multiple Alternative Company-Wide Compliance Plans.

I.D.5.c.(ii) The Compliance Plan must be submitted on the Division-approved form and include all of the following elements:

I.D.5.c.(ii)(A) A list of all of the engines that will rely on this Section I.D.5.c. to comply with the standards established in Section I.D.5.b. Each engine must be identified by AIRS number, location (inside or outside the 8-Hour Ozone Control Area and facility name), horsepower, manufacturer, model and serial number, historical annual operating hours (averaged over 2017, 2018, and 2019), and engine configuration.

I.D.5.c.(ii)(B) For each engine included in the Alternative Company-Wide Compliance Plan:

I.D.5.c.(ii)(B)(1) Identification of the most stringent NO_x emission standard (in g/hp-hr or converted to g/hp-hr, if not expressed as such in the applicable permit) and operating conditions applicable to the engine under any rule or permit condition in effect on or before November 14, 2020.

I.D.5.c.(ii)(B)(2) Identification of the g/hp-hr at which the engine is operating on or before November 14, 2020, if different than the rate identified in Section I.D.5.c.(ii)(B)(1).

- I.D.5.c.(ii)(B)(3) The emission standards (in g/hp-hr) and any operating conditions with which each engine will comply under the Alternate Company-Wide Compliance Plan, including any intended shut-downs, including any modifications or changes made to comply with the VOC or CO standards in Table 2.
- I.D.5.c.(ii)(B)(4) The date by which each engine will meet the emission standards or other operating conditions identified in Section I.D.5.c.(ii)(B)(3), consistent with Table 4.
- I.D.5.c.(ii)(B)(5) The maximum allowable NOx emissions (in tons/year) based on limits applicable on or before November 14, 2020, as identified in Section I.D.5.c.(ii)(B)(1).
- I.D.5.c.(ii)(B)(6) The historic NOx emissions (in tons/year) averaged over calendar years 2017, 2018 and 2019, based on actual operating hours and permitted emission standards.
- I.D.5.c.(ii)(B)(7) The NOx emissions that would be allowed on an annual basis (in tons/year) assuming the engine was complying with the emission standards established in Table 2.
- I.D.5.c.(ii)(B)(8) Each engine's allowable NOx emissions (in tons/year) when operated in accordance with limitations identified in Section I.D.5.c.(ii)(B)(3), including any increase in NOx emissions that result from modifications or changes made to comply with the VOC or CO standards in Table 2.
- I.D.5.c.(ii)(C) The total allowable NOx emissions (in tons/year) calculated for all engines in the Alternative Company-Wide Compliance Plan, as specified in Section I.D.5.c.(ii)(B)(5).

I.D.5.c.(ii)(D) The total NOx emissions (in tons/year) calculated for all engines in the Alternative Company-Wide Compliance Plan, as specified in Section I.D.5.c.(ii)(B)(6).

I.D.5.c.(ii)(E) The total NOx emissions calculated for all engines included in the Alternate Company-Wide Compliance Plan assuming all engines were complying with the emission standards established in Table 2, as specified in Section I.D.5.c.(ii)(B)(7).

I.D.5.c.(ii)(F) The total allowable NOx emissions (in tons/year) calculated for all engines included in the Alternate Company-Wide Compliance Plan, as specified in Section I.D.5.c.(ii)(B)(8).

I.D.5.c.(ii)(G) A calculation of:

I.D.5.c.(ii)(G)(1) The difference between Section I.D.5.c.(ii)(C) and Section I.D.5.c.(ii)(F). This difference is called the “Plan Emission Reductions”.

I.D.5.c.(ii)(G)(2) The difference between total historic NOx emissions as calculated in Section I.D.5.c.(ii)(D) and the total allowable NOx emissions (in tons/year) for all engines included in the Alternate Company-Wide Compliance Plan assuming all engines were complying with Table 2, as specified in Section I.D.5.c.(ii)(E).

I.D.5.c.(ii)(G)(3) The difference between total historic NOx emissions as calculated in Section I.D.5.c.(ii)(D) and the total allowable NOx emissions (in tons/year) for all engines included in the Alternate Company-Wide Compliance Plan, as specified in Section I.D.5.c.(ii)(F).

I.D.5.c.(ii)(H) A demonstration that:

- I.D.5.c.(ii)(H)(1) The total NO_x emissions allowed under the Alternative Company-Wide Compliance Plan (Section I.D.5.c.(ii)(F)) are less than or equal to the total NO_x emissions that would be allowed under Table 2 (Section I.D.5.c.(ii)(E)).
- I.D.5.c.(ii)(H)(2) The reductions from emissions achieved by the Alternative Company-Wide Compliance Plan are greater than or equal to the reductions from actual emissions achieved by Table 2 (i.e. that the figure calculated in Section I.D.5.c.(ii)(G)(3) is greater than or equal to the figure calculated in Section I.D.5.c.(ii)(G)(2)).
- I.D.5.c.(ii)(I) A certification by the owner or operator that based on information and belief formed after reasonable inquiry, the statements and information in the Compliance Plan are true, accurate, and complete.
- I.D.5.c.(iii) Any owner or operator utilizing this Alternative Company-Wide Compliance Plan must meet the emission standards for NO_x, CO and VOC as identified in I.D.5.c.(ii)(B)(3) by the compliance deadlines listed in Table 4.
- I.D.5.c.(iv) Owners and operators must modify existing permits to reflect the emission standards or other operating conditions identified in the Compliance Plan (Section I.D.5.c.(ii)(B)(3)) for that engine. Permit applications must be submitted to the Division at least 365 days prior to the date established in Section I.D.5.c.(ii)(B)(4) for that engine.
- I.D.5.c.(v) Compliance Plan Updates. By May 1st of each year (beginning in 2022) and continuing through and including the final year of a Compliance Plan, an owner or operator must submit an update to the Compliance Plan with the following information:

- I.D.5.c.(v)(A) For each engine, any change in location and any action taken under the Compliance Plan (e.g., permit modification applied for, engine retrofit completed, engine taken offline) and the date;
- I.D.5.c.(v)(B) A calculation of the percentage of Plan Emission Reductions achieved as of the date of submittal of the update (in each compliance period and cumulatively);
- I.D.5.c.(v)(C) Any changes made to the Compliance Plan (e.g. change in compliance date for an engine). No change to the compliance date for an engine can be made after the date established in the Compliance Plan for that engine.;
- I.D.5.c.(v)(D) If ownership or operation of an engine in the Compliance Plan for which emission reductions were included in the calculation of Plan Emission Reductions was sold or transferred in the previous year, an identification of how the owner or operator will achieve the portion of Plan Emission Reductions attributed to that engine under the Compliance Plan (the difference between Section I.D.5.c.(ii)(B)(5) and (8)).
- I.D.5.c.(v)(E) A certification by the owner or operator that based on information and belief formed after reasonable inquiry, the statements and information in the update are true, accurate, and complete.
- I.D.5.c.(vi) Nothing in this Section I.D.5.c exempts an engine that is part of an Alternative Company-Wide Compliance Plan from compliance with the performance testing, monitoring, recordkeeping or reporting requirements of this Section I.D.5.

TABLE 4					
Location of Subject Engines covered by the Alternative Company-Wide Compliance Plan	Compliance Deadlines				
	May 1, 2022	May 1, 2023	May 1, 2024	May 1, 2025	May 1, 2026
	Percent (%) of Plan Emission Reductions Achieved				
Inside, or inside and outside, the 8-Hour Ozone Control Area	At least 50% of Plan Emission Reductions from engines inside the 8-Hour Ozone Control Area	At least 75% of Plan Emission Reductions from engines inside the 8-Hour Ozone Control Area; and at least 25% of Plan Emission Reductions from engines outside the 8-Hour Ozone Control Area	100% of Plan Emission Reductions from engines inside the 8-Hour Ozone Control Area; and at least 50% of Plan Emission Reductions from engines outside the 8-Hour Ozone Control Area	At least 75% of Plan Emission Reductions from engines outside the 8-Hour Ozone Control Area	100% of Plan Emission Reductions
Outside the 8-Hour Ozone Control Area only	At least 20%	At least 40%	At least 60%	At least 80%	100%

I.D.5.d. Performance Testing

I.D.5.d.(i) Engines subject to this Section I.D.5. must conduct a performance test consistent with the requirements of this Section I.D.5.d.

I.D.5.d.(i)(A) The owner or operator of an engine subject to Section I.D.5.b.(ii) must conduct a performance test for NO_x, CO, and O₂ by May 1, 2021.

I.D.5.d.(i)(B) The owner or operator of an engine placed in service, modified, relocated or replaced after May 1, 2021 must conduct a performance test within 12 months of the date the engine is placed in service, modified, relocated or replaced.

I.D.5.d.(i)(C) The following engines are exempt from the requirements of this Section I.D.5.d.

I.D.5.d.(i)(C)(1) Engines subject to the performance testing requirements of 40 C.F.R. Part 60, Subpart JJJJ (July 1, 2019).

I.D.5.d.(i)(C)(2) Engines subject to at least semi-annual portable analyzer testing or ongoing performance testing in a permit issued on or before November 14, 2020.

I.D.5.d.(i)(D) A performance test conducted in accordance with 40 C.F.R. §60.4244 (July 1, 2019) between January 1, 2020 and May 1, 2021 will satisfy the initial performance testing requirements in Section I.D.5.d.(i)(A).

I.D.5.d.(ii) Performance tests must be conducted in accordance with the applicable reference test methods of 40 C.F.R. Part 60, Appendix A (July 1, 2019), and a test protocol submitted to the Division for review at least thirty (30) days prior to testing and in accordance with AQCC Common Provisions Regulation Section II.C.

I.D.5.d.(iii) Tuning of an engine prior to the performance test required by this Section I.D.5.d is not a violation of this rule. However, readjustment of an engine set point following the performance test that would negatively impact the performance of the engine (i.e. result in increased emissions above applicable permit limits) is a violation of this rule.

I.D.5.e. Monitoring. Except as provided in Section I.D.5.a.(ii), owners or operators of an engine subject to Section I.D.5.a must:

I.D.5.e.(i) Beginning on May 1, 2022, conduct semi-annual portable analyzer monitoring for NO_x, CO, and O₂. At least one calendar month must separate the semi-annual tests.

I.D.5.e.(i)(A) If the engine is operated for less than 200 hrs in any semi-annual period, then the portable analyzer monitoring need not occur during that semi-annual period (i.e. the engine does not need to be started for the sole purpose of portable monitoring).

I.D.5.e.(i)(B) All portable analyzer testing required by this section must be conducted using the Division's Portable Analyzer Monitoring Protocol (version: March 2006).

I.D.5.e.(i)(C) Tuning of an engine prior to semi-annual monitoring events required by this Section I.D.5.e.(i) is not a violation of this rule. However, readjustment of an engine set point following the monitoring event that would negatively impact the performance (i.e. result in increased emissions above applicable permit limits) of the engine is a violation of this rule.

I.D.5.e.(i)(D) A performance test conducted pursuant to Section I.D.5.d., 40 C.F.R. Part 60, JJJJ, or a permit requirement may take the place of the next required semi-annual portable analyzer test required by this section.

I.D.5.e.(i)(E) An engine subject to at least semi-annual portable analyzer testing requirements in an existing permit issued by the Division can comply with this Section I.D.5.e.(i) by complying with the testing requirements in the permit.

- I.D.5.e.(ii) Beginning May 1, 2021, if a catalyst is used to reduce emissions:
 - I.D.5.e.(ii)(A) Monitor the inlet temperature to the catalyst daily and conduct maintenance if the temperature is not within applicable limits.
 - I.D.5.e.(ii)(B) Measure the pressure drop across the catalyst monthly and conduct maintenance if the pressure drop is greater than 2 inches outside the baseline value established after each semi-annual portable analyzer monitoring.
 - I.D.5.e.(ii)(C) Engines that are subject to catalyst temperatures and catalyst pressure drop monitoring requirements in an existing permit issued by the Division or 40 C.F.R. Part 63, Subpart ZZZZ (July 1, 2019) satisfy the monitoring requirements of this Section I.D.5.e.(ii).
- I.D.5.e.(iii) Beginning May 1, 2021 or the date the engine is placed in service, modified, relocated or replaced (if later), install (if not already) and operate an hour meter or Division approved alternate method to continuously track the hours of operation of the subject engine.
- I.D.5.e.(iv) Conduct the following inspections and adjustments at least annually, unless otherwise specified, beginning in 2022
 - I.D.5.e.(iv)(A) Change oil and filters as necessary; and,
 - I.D.5.e.(iv)(B) Inspect air cleaners, fuel filters, hoses, and belts and clean or replace as necessary; and,
 - I.D.5.e.(iv)(C) Inspect spark plugs and replace as necessary; or,

I.D.5.e.(iv)(D) Conduct a combustion process adjustment according to the manufacturer recommended procedures and schedule. Alternatively, the owner or operator may rely on a combustion process adjustment conducted in accordance with requirements and schedules of a New Source Performance Standard in 40 CFR Part 60 (July 1, 2022) or National Emission Standard for Hazardous Air Pollutants in 40 CFR Part 63 (July 1, 2022) conducted during the same annual period to satisfy the annual combustion process adjustment requirement of this Section I.D.5.e.(iv)(D) for that 12-month period.

I.D.5.f.Recordkeeping. The following records must be kept for a period of five years and made available to the Division upon request.

I.D.5.f.(i) Records of performance tests conducted pursuant to Section I.D.5.d, including I.D.5.d.(i)(D)., including the date, engine settings on the date of the test, and documentation of the methods and results of the testing.

I.D.5.f.(ii) Records of semi-annual portable analyzer monitoring, including the date, engine settings on the date of the monitoring, and documentation of the results of the monitoring. These records must include any demonstration that no semi-annual portable analyzer monitoring was required as provided under Section I.D.5.e.(i)(D) or I.D.5.e.(i)(E), if applicable.

I.D.5.f.(iii) Records of catalyst monitoring required by Section I.D.5.e.(ii) and any actions taken to address monitored values outside the temperature or pressure drop parameters, including the date and a description of actions taken.

I.D.5.f.(iv) If claiming an exemption under Section I.D.5.a.(ii), records demonstrating that fuel combustion was less than 100 MMBtu per year or hours of operation are less than 250 hours per year.

I.D.5.f.(v) Hours of operation as recorded by the hour meter or alternative device approved by the Division continuously tracking hours as required by Section I.D.5.e.(iii), at least on a calendar month basis.

I.D.5.f.(vi) Records of tuning, adjustments, or other combustion process adjustments required under Section I.D.5.e.(iv), including:

I.D.5.f.(vi)(A) The date of the adjustment.

I.D.5.f.(vi)(B) A description of any corrective action taken.

I.D.5.f.(vi)(C) If the owner or operator conducts the combustion process adjustment according to the manufacturer recommended procedures and schedule and the manufacturer specifies a combustion process adjustment on an operation time schedule, the hours of operation since the last combustion process adjustment and the procedures followed. The owner or operator must retain documentation of any relied upon manufacturer recommended procedures, specifications, and maintenance schedule for five years after the owner or operator ceases to rely upon it.

I.D.5.f.(vi)(D) If the owner or operator conducts the combustion process adjustment according to a New Source Performance Standard or National Emission Standard for Hazardous Air Pollutants, what standard applied and what procedures were followed.

I.D.5.g. Reporting. Beginning on the date specified and by May 1 of each year thereafter, the owner or operator of each engine subject to this Section I.D.5. must submit the following information covering the preceding calendar year:

I.D.5.g.(i) Beginning May 1, 2021, a statement of the status of performance testing required under Section I.D.5.d, and the date and results of that testing;

II.D.5.g.(ii) Beginning May 1, 2022, an identification of any engines placed in service, modified, relocated, or replaced, including AIRS number, serial number, location, engine configuration, and a certification as to whether the emission standards in Table 2 are met;

I.D.5.g.(iii) Beginning May 1, 2023, the date on which the monitoring required by Sections I.D.5.e.(iv) was performed;

I.D.5.g.(iv) Beginning May 1, 2023, the date that all required semi-annual portable analyzer testing was performed under Section I.D.5.e.(i), and the results of that testing.

I.D.6. (State Only) Additional Requirements for Internal Combustion Engines

I.D.6.a. Applicability

I.D.6.a.(i) Sections I.D.6.a. through I.D.6.f.(i) apply to stationary rich burn natural gas fired reciprocating internal combustion engines state-wide with a manufacturer's design rate greater than or equal to 100 horsepower but less than 1000 horsepower and lean burn natural gas fired reciprocating internal combustion engines state-wide with a manufacturer's design rate greater than or equal to 250 horsepower but less than 1000 horsepower.

I.D.6.a.(ii) Sections I.D.6. through I.D.6.f.(i) apply to stationary diesel or dual-fuel fired internal combustion engines state-wide with a manufacturer's design rate greater than or equal to 500 horsepower.

I.D.6.a.(iii) For purposes of this Section I.D.6., modified means any physical change to the engine or change in method of operation that results in an increase in the emission rate of any air pollutant, and does not include any physical or operational changes excluded by 40 C.F.R. 60.14(e).

I.D.6.a.(iv) For purposes of this Section I.D.6., placed in service means the bringing of an engine on-site for use. The placed in service date is the date the engine begins to operate. For engines in service on or before January 30, 2024, replacement under an authorized alternative operating scenario, or a one-time only replacement of an engine that cannot achieve an applicable emission standard due to technical or economic infeasibility, is not considered placed in service.

I.D.6.a.(v) For purposes of this Section I.D.6., relocated means (1) the bringing of an engine into the 8-hour ozone control area or Northern Weld County from outside the 8-hour ozone control area or Northern Weld County, or (2) the bringing of an engine into the State of Colorado from outside the State of Colorado. The relocation date is the date the subject engine begins to operate.

I.D.6.a.(vi) Exemptions.

I.D.6.a.(vi)(A) Engines that burn less than 100 MMBtu per year of natural gas or diesel fuel on a rolling-12-month basis are not subject to Sections I.D.6.a. through I.D.6.d, or I.D.6.f.

I.D.6.a.(vi)(B) Non-road engines, as defined in Regulation Number 3, Part A, Section I.B.36 are not subject to this Section I.D.6.

I.D.6.a.(vi)(C) Any emergency power generator exempt from APEN or construction permit requirements pursuant to Regulation Number 3, Parts A or B are not subject to this Section I.D.6.

I.D.6.a.(vi)(D) Emergency power generators or engines that are used to meet electrical demand or provide voltage support to the electrical grid that operate less than 250 hours per year on a rolling-12-month basis are not subject to Sections I.D.6.a. through I.D.6.d, or I.D.6.f.

I.D.6.a.(vi)(E) Stationary natural gas-fired reciprocating internal combustion engines placed in service, modified, or relocated before January 30, 2024, if the owner or operator will remove or electrify the engine as an emissions reduction measure to comply with Regulation Number 7, Part B, Section VII.

I.D.6.b. Emission Standards for Engines Subject to Section I.D.6.a.

I.D.6.b.(i) The owner or operator of any stationary natural gas-fired reciprocating internal combustion engine that is placed in service, modified, or relocated after January 30, 2024, must comply with the emission standards in Table 5 upon placement in service, modification, or relocation.

I.D.6.b.(ii) The owner or operator of any stationary natural gas-fired reciprocating internal combustion engine not subject to Section I.D.6.b.(i) must comply with the emission standards in Table 5 in accordance with the timing set forth Section I.D.6.b.(vii).

TABLE 5	
Engine Type	Emission Standards
	NOx (g/hp-hr)
Lean Burn engines in service on or before January 30, 2024	0.8
Lean Burn <u>engines placed in service, modified, or relocated</u> after January 30, 2024	0.5
Rich Burn <u>engines</u>	0.5

I.D.6.b.(iii) The owner or operator of any stationary diesel or dual-fuel internal combustion engine that is placed in service, modified, or relocated after January 30, 2024, must achieve or exceed EPA Tier IV standards for NOx upon placement in service, modification, or relocation.

I.D.6.b.(iv) The owner or operator of any stationary diesel or dual-fuel internal combustion engine not subject to Section I.D.6.b.(iii) must achieve or exceed EPA Tier IV standards for NOx in accordance with the timing set forth Section I.D.6.b.(vii).

I.D.6.b.(v) The owner or operator of any engine that meets all of the criteria described in Sections I.D.6.b.(v)(A) through (E) may submit a request to the Division for an alternative emission standard for a specific engine based on technical or economic infeasibility. To qualify for an alternative emission standard, an owner or operator must submit to the Division a reasonable demonstration detailing why it is not technically or economically feasible for the individual engine to achieve the emission standard in Section I.D.6.b.(ii) by January 30, 2025. Within a reasonable amount of time after receipt, the Division will grant an alternative emission standard where the Division determines a satisfactory source-specific demonstration has been made. The Division will set an appropriate alternative emission standard considering control technology that will achieve the maximum degree of emission control that a particular source is capable of meeting and that is reasonably available considering technological and economic feasibility. Any granted alternative emission standard will be included in the applicable permit pursuant to Section I.D.6.b.(vi)

I.D.6.b.(v)(A) Stationary natural gas-fired lean burn reciprocating internal combustion engine;

I.D.6.b.(v)(B) Located outside northern Weld County and the 8-Hour Ozone Control Area;

I.D.6.b.(v)(C) Placed in service on or before January 30, 2024;

I.D.6.b.(v)(D) With greater than or equal to 400 horsepower; and

I.D.6.b.(v)(E) With a NOx emission factor of less than or equal to 3.0 g/hp-hr as reflected by the APEN on file as of January 30, 2024.

I.D.6.b.(vi) Permit Modification.

I.D.6.b.(vi)(A) An engine in service on or before January 30, 2024, that requires only a modification of an existing permit to meet the emission standards in this Section I.D.6.b. must submit a complete permit application containing the necessary limitations by the following deadlines.

I.D.6.b.(vi)(A)(1) For individual construction permit or Title V operating permit significant modification applications, by no later than May 1, 2024.

I.D.6.b.(vi)(A)(2) For general construction permit or Title V operating permit minor modification applications, by no later than January 30, 2025.

I.D.6.b.(vi)(B) For any engine in service on or before January 30, 2024, and not subject to Section I.D.6.b.(vi)(A), owners and operators must modify existing permits to reflect the emission standards or other operating conditions necessary to achieve compliance with Table 5. Complete permit applications must be submitted to the Division by the following deadlines:

I.D.6.b.(vi)(B)(1) For individual construction permit or Title V operating permit modifications applications, at least 365 days prior to the date established in Section I.D.6.(vii) for that engine; and

I.D.6.b.(vi)(B)(2) For general construction permit modification applications, at least 90 days prior to the date established in Section I.D.6.b.(vii) for that engine.

I.D.6.b.(vii) Compliance deadlines for engines subject to Sections I.D.6.b.(ii) or I.D.6.b.(iv).

I.D.6.b.(vii)(A) Engines that comply with the emission standards on or before January 30, 2024, or are subject to Section I.D.6.b.(vi)(A) must meet the emission standards in Table 5 by May 1, 2025. If an owner or operator has submitted a complete application in compliance with Section I.D.6.b.(vi)(A) and has not been issued the relevant permit by May 1, 2025, the Division may extend the compliance deadline for an engine as necessary to align with issuance date of the permit.

I.D.6.b.(vii)(B) Engines not subject to Section I.D.6.b.(vi)(A) must meet the emission standards in Table 5 in accordance with the timing set forth in Table 6. If an owner or operator has submitted a complete application requesting the emission standards or other operating conditions necessary to achieve compliance with Table 5 and has not been issued the relevant permit by the applicable compliance deadline in Table 6, the Division may extend the compliance deadline for an engine as necessary to align with issuance date of the permit.

TABLE 6					
Location of Subject Engines by Owner or Operator	Compliance Deadlines				
	May 1, 2025	May 1, 2026	May 1, 2027	May 1, 2028	May 1, 2029
	Percent (%) of engines that must comply with Table 5 limits				
Natural Gas-Fired Engines Inside, or inside and outside, the 8-Hour Ozone Control Area	At least 34% of engines inside the 8-Hour Ozone Control Area	At least 67% of engines inside the 8-Hour Ozone Control Area; and at least 25% of engines outside the 8-Hour Ozone Control Area	100% of engines in the 8-Hour Ozone Control Area; and at least 50% of engines outside the 8-Hour Ozone Control Area	At least 75% of engines outside the 8-Hour Ozone Control Area	100% of all engines
Natural Gas-Fired Engines Outside the 8-Hour Ozone Control Area only	At least 20%	At least 40%	At least 60%	At least 80%	100%
Diesel and Dual Fuel Engines State-wide			At least 50%		100%

I.D.6.b.(viii) If an owner or operator replaces an engine subject to an emission standard under this Section I.D.6.b. with a different stationary engine under an alternative operating scenario pursuant to an existing Division issued permit, the replacement engine must, meet the same emission standard as the engine being replaced.

I.D.6.c. Performance Testing

I.D.6.c.(i) Engines subject to this Section I.D.6. must conduct a performance test consistent with the following requirements.

I.D.6.c.(i)(A) The owner or operator of an engine subject to Section I.D.6.b.(ii) must conduct a performance test for NOx by May 1, 2025 or within 6 months of any Division established compliance deadline pursuant to Section I.D.6.b.(vii).

I.D.6.c.(i)(B) The owner or operator of an engine placed in service, modified, relocated or replaced after January 30, 2024, must conduct a performance test within 12 months of the date the engine is placed in service, modified, relocated or replaced.

I.D.6.c.(i)(C) The following engines are exempt from the requirements of this Section I.D.6.c.

I.D.6.c.(i)(C)(1) Engines subject to the performance testing requirements of 40 C.F.R. Part 60, Subpart JJJJ (July 1, 2023) or 40 C.F.R. Part 60, Subpart IIII (July 1, 2023).

I.D.6.c.(i)(C)(2) Engines subject to at least semi-annual portable analyzer testing or ongoing performance testing in a permit issued on or before January 30, 2024.

I.D.6.c.(i)(D) A performance test conducted in accordance with 40 C.F.R. §60.4244 (July 1, 2023) or 40 C.F.R. §60.4212 or 60.4213 (July 1, 2023) between January 1, 2023 and May 1, 2025, will satisfy the initial performance testing requirements in Section I.D.6.c.(i)(A).

I.D.6.c.(ii) Performance tests must be conducted in accordance with the applicable reference test methods of 40 C.F.R. Part 60, Appendix A (July 1, 2023), and a test protocol submitted to the Division for review at least thirty (30) days prior to testing and in accordance with AQCC Common Provisions Regulation Section II.C.

I.D.6.c.(iii) Tuning of an engine prior to the performance test required by this Section I.D.6.c. is not a violation of this rule. However, readjustment of an engine set point following the performance test that would negatively impact the performance of the engine (i.e., result in increased emissions above applicable permit limits) is a violation of this rule.

I.D.6.c.(iv) Diesel engines covered by Section I.D.6.b.(iii) or (iv) that are certified by the manufacturer as Tier IV engines do not require the initial performance test under Section I.D.6.c.(i)(B), but must conduct annual performance testing beginning five years after the engine is placed in service.

I.D.6.d. Monitoring. Except as provided in Section I.D.6.a.(vi), owners or operators of an engine subject to Section I.D.6.a. must:

I.D.6.d.(i) Beginning on May 1, 2025 or within 6 months of any Division established compliance deadline pursuant to Section I.D.6.b.(vii)., conduct annual portable analyzer monitoring for NOx. At least six calendar months must separate the annual tests.

I.D.6.d.(i)(A) If the engine is operated for less than 200 hours in any 12-month period, then the portable analyzer monitoring need not occur during that annual period (i.e., the engine does not need to be started for the sole purpose of portable monitoring).

I.D.6.d.(i)(B) All portable analyzer testing required by this section must be conducted using the Division's Portable Analyzer Monitoring Protocol (version: March 2006).

I.D.6.d.(i)(C) Tuning of an engine prior to annual monitoring events required by this Section I.D.6.d.(i) is not a violation of this rule. However, readjustment of an engine set point following the monitoring event that would negatively impact the performance (i.e., result in increased emissions above applicable permit limits) of the engine is a violation of this rule.

- I.D.6.d.(i)(D) A performance test conducted pursuant to Section I.D.6.c., 40 C.F.R. Part 60, JJJJ, IIII, or a permit requirement may take the place of the next required annual portable analyzer test required by this section.
- I.D.6.d.(i)(E) An engine subject to at least annual portable analyzer testing requirements in an existing permit issued by the Division can comply with this Section I.D.6.d.(i) by complying with the testing requirements in the permit.
- I.D.6.d.(ii) For natural gas-fired engines, beginning May 1, 2024, if a catalyst is used to reduce emissions:
 - I.D.6.d.(ii)(A) Monitor the inlet temperature to the catalyst weekly and conduct maintenance if the temperature is not within applicable limits.
 - I.D.6.d.(ii)(B) Measure the pressure drop across the catalyst monthly and conduct maintenance if the pressure drop is greater than 2.0 inches of water outside the baseline value established after each annual portable analyzer monitoring.
 - I.D.6.d.(ii)(C) Engines that are subject to catalyst temperatures and catalyst pressure drop monitoring requirements in an existing permit issued by the Division or 40 C.F.R. Part 63, Subpart ZZZZ (July 1, 2023) satisfy the monitoring requirements of this Section I.D.6.d.(ii).
- I.D.6.d.(iii) Conduct the following inspections and adjustments at least annually, unless otherwise specified, beginning in 2025.
 - I.D.6.d.(iii)(A) Change oil and filters as necessary; and,
 - I.D.6.d.(iii)(B) Inspect air cleaners, fuel filters, hoses, and belts and clean or replace as necessary; and,
 - I.D.6.d.(iii)(C) Inspect spark plugs and replace as necessary; or,

I.D.6.d.(iii)(D) Conduct a combustion process adjustment according to the manufacturer recommended procedures and schedule. Alternatively, the owner or operator may rely on a combustion process adjustment conducted in accordance with requirements and schedules of a New Source Performance Standard in 40 CFR Part 60 (July 1, 2023) or National Emission Standard for Hazardous Air Pollutants in 40 CFR Part 63 (July 1, 2023) conducted during the same annual period to satisfy the annual combustion process adjustment requirement of this Section I.D.6.d.(iv)(D) for that 12-month period.

I.D.6.e. Recordkeeping. The following records must be kept for a period of five years and made available to the Division upon request.

I.D.6.e.(i) Records of performance tests conducted pursuant to Section I.D.6.c., including I.D.6.c.(i)(D), including the date, engine settings on the date of the test, and documentation of the methods and results of the testing.

I.D.6.e.(ii) Records of annual portable analyzer monitoring, including the date, engine settings on the date of the monitoring, and documentation of the results of the monitoring. These records must include any demonstration that no annual portable analyzer monitoring was required as provided under Sections I.D.6.d.(i)(D) or I.D.6.d.(i)(E), if applicable.

I.D.6.e.(iii) For natural gas-fire engines, records of catalyst monitoring required by Section I.D.6.d.(ii) and any actions taken to address monitored values outside the temperature or pressure drop parameters, including the date and a description of actions taken.

I.D.6.e.(iv) If claiming an exemption under Section I.D.6.a.(vi), records demonstrating that fuel combustion was less than 100 MMBtu per year or hours of operation are less than 250 hours per year.

I.D.6.e.(v) Hours of operation at least on a calendar month basis.

I.D.6.e.(vi) Records of tuning, adjustments, or other combustion process adjustments required under Section I.D.6.d.(iii), including:

I.D.6.e.(vi)(A) The date of the adjustment.

I.D.6.e.(vi)(B) A description of any corrective action taken.

I.D.6.e.(vi)(C) If the owner or operator conducts the combustion process adjustment according to the manufacturer recommended procedures and schedule and the manufacturer specifies a combustion process adjustment on an operation time schedule, the hours of operation since the last combustion process adjustment and the procedures followed. The owner or operator must retain documentation of any relied upon manufacturer recommended procedures, specifications, and maintenance schedule for five years after the owner or operator ceases to rely upon it.

I.D.6.e.(vi)(D) If the owner or operator conducts the combustion process adjustment according to a New Source Performance Standard or National Emission Standard for Hazardous Air Pollutants, what standard applied and what procedures were followed.

I.D.6.f. Reporting.

I.D.6.f.(i) Beginning on the date specified and by May 1 of each year thereafter, the owner or operator of each engine subject to this Section I.D.6. must submit the following information covering the preceding calendar year.

I.D.6.f.(i)(A) Beginning May 1, 2024, a statement of the status of performance testing required under Section I.D.6.c., and the date and results of that testing (i.e., pass or fail).

II.D.6.f.(i)(B) Beginning May 1, 2025, an identification of any engines placed in service, modified, relocated, or replaced, including AIRS number, serial number, location, engine configuration, and a certification as to whether the emission standards in Table 5 are met.

I.D.6.f.(i)(C) Beginning May 1, 2026, the date on which the monitoring required by Sections I.D.6.d.(iii) was performed.

I.D.6.f.(i)(D) Beginning May 1, 2026, the date that all required annual portable analyzer testing was performed under Section I.D.6.d.(i), and the results of that testing (i.e., pass or fail).

II. Control of Emissions from Stationary and Portable Combustion Equipment in the 8-Hour Ozone Control Area or Northern Weld County

II.A. Requirements for major sources of NO_x

II.A.1. Applicability.

II.A.1.a. Except as provided in Section II.A.2., the requirements of this Section II. apply to owners and operators of any stationary combustion equipment that existed at a major source of NO_x (greater than or equal to 100 tpy NO_x) as of June 3, 2016, located in the 8-Hour Ozone Control Area.

II.A.1.b. Except as provided in Section II.A.2., the requirements of Section II. apply to owners and operators of any stationary combustion equipment that existed at a major source of NO_x (greater than or equal to 50 tpy NO_x) as of January 27, 2020, located in the 8-Hour Ozone Control Area, that is not already subject as provided under Section II.A.1.a.

II.A.1.c. Except as provided in Section II.A.2., the requirements of Section II. apply to owners and operators of process heaters that existed at source that emits, or has the potential to emit, NO_x emissions greater than or equal to 25 tpy NO_x as of July 20, 2021, located in the 8-Hour Ozone Control Area, that is not already subject as provided under Sections II.A.1.a. or II.A.1.b.

II.A.1.d. Except as provided in Section II.A.2., the requirements of Section II. apply to owners and operators of any stationary combustion equipment that existed at a major source of NO_x (greater than or equal to 25 tpy NO_x) as of November 7, 2022, located in the 8-Hour Ozone Control Area, that is not already subject as provided under Sections II.A.1.a. through II.A.1.c.

II.A.1.e. Except as provided in Sections II.A.2. or III.C., the requirements of Section II. apply to owners and operators of any stationary combustion equipment that existed at a major source of NO_x (greater than or equal to 100 tpy NO_x) as of November 7, 2022, located in northern Weld County.

II.A.1.f. Except as provided in Section II.A.2., the requirements of Section II. apply to owners and operators of any stationary combustion equipment that existed at a major source of NO_x (greater than or equal to 50 tpy NO_x) as of July 24, 2024, located in northern Weld County, that is not already subject as provided under Section II.A.1.e.

II.A.2. Exemptions. The following stationary combustion equipment are exempt from the emission limitation requirements of Section II.A.4., the compliance demonstration requirements in Section II.A.5., and the related recordkeeping and reporting requirements of Sections II.A.7.a through II.A.7.e. and II.A.8, but these sources must maintain any and all records necessary to demonstrate that an exemption applies. These records must be maintained for a minimum of five years and made available to the Division upon request. Qualifying for an exemption in this section does not preclude the combustion process adjustment requirements of Section II.A.6., when required by II.A.6.a.

Once stationary combustion equipment no longer qualifies for any exemption, the owner or operator must comply with the applicable requirements of this Section II.A. as expeditiously as practicable but no later than 36 months after any exemption no longer applies. Additionally, once stationary combustion equipment that is not equipped with CEMS or CERMS no longer qualifies for any exemption, the owner or operator must conduct a performance test using EPA test methods within 180 days and notify the Division of the results and whether emission controls will be required to comply with the emission limitations of Section II.A.4.

II.A.2.a. Any stationary combustion equipment whose utilization is less than:

II.A.2.a.(i) 20% of its capacity factor on an annual average basis over a 3-year rolling period for boilers; or

II.A.2.a.(ii) 10% of its capacity factor on an annual average basis over a 3-year rolling period for stationary combustion turbines and compression ignition reciprocating internal combustion engines.

II.A.2.b. An engine testing operation or process line.

II.A.2.c. Any gaseous fuel fired stationary combustion equipment used to control VOC emissions from a commercial or industrial process.

II.A.2.d. Any stationary combustion equipment with total uncontrolled actual emissions less than 5 tpy NO_x on a calendar year basis.

II.A.2.e. Any natural gas-fired reciprocating internal combustion engines subject to a work practice or emission control requirement contained in this Regulation Number 26, Part B, Sections I.A. or I.B.

II.A.2.f. Any stationary combustion equipment subject to a federally enforceable work practice or emission control requirement contained in this Regulation Number 26, Part B, Sections III.A. through III.C. or Regulation 23.

II.A.3. Definitions

II.A.3.a. "Affected unit" means any stationary combustion equipment that is subject to or becomes subject to an emission limitation in Section II.A.4.

II.A.3.b. "Boiler" means an enclosed device using controlled flame combustion and having the primary purpose of recovering thermal energy in the form of steam or hot water.

- II.A.3.c. “Capacity factor” means the ratio of the amount of fuel burned by an emissions unit in a calendar year to the amount of fuel it could have burned if it had operated at the designed heat input rating for 8,760 hours during the calendar year. Alternatively, for electric generating units, capacity factor can mean the ratio of the unit’s actual annual electric output (expressed in MWe/hr) to the electric output the unit could have achieved if it operated at its nameplate capacity (or maximum observed hourly gross load (expressed in MWe/hr) if greater than the nameplate capacity) for 8,760 hours during the calendar year.
- II.A.3.d. “Ceramic kiln” means equipment used for the curing or firing of ceramic products or glaze on ceramic products. A kiln may operate continuously or by batch process.
- II.A.3.e. “Continuous emission monitoring system” (“CEMS”) or “Continuous emission rate monitoring system” (“CERMS”) means the total equipment required to sample, condition (if applicable), analyze, and provide a written record of such emissions and/or emission rates, expressed on a continuous basis in terms of an applicable emission limitation. Such equipment includes, but is not limited to, sample collection and calibration interfaces, pollutant analyzers, a diluent analyzer (oxygen or carbon dioxide), stack gas volumetric flow monitors (if appropriate for CERMS), and data recording and storage devices.
- II.A.3.f. “Compression ignition reciprocating internal combustion engine (RICE)” means a type of stationary RICE that is liquid fuel-fired and not ignited with a spark plug or other sparking device.
- II.A.3.g. “Digester gas” means any gaseous byproduct of wastewater treatment typically formed through the anaerobic decomposition of organic waste materials and composed principally of methane and carbon dioxide.
- II.A.3.h. “Duct burner” means a device that combusts fuel and is placed in the exhaust duct from another source (e.g., stationary combustion turbine, internal combustion engine, or kiln) to allow the firing of additional fuel to heat the exhaust gases before the exhaust gases enter a heat recovery steam generating unit.
- II.A.3.i. “Dryer” means a device that is used to reduce or evaporate moisture content or remove organic contaminants.

- II.A.3.j. “Furnace” means an enclosed device that is an integral component of a manufacturing process and that uses thermal treatment to accomplish recovery of materials or energy.
- II.A.3.k. “Gaseous fuel” means natural gas, landfill gas, refinery fuel gas, digester gas, methane, ethane, propane, butane, or any gas stored as a liquid at high pressure such as liquefied petroleum gas.
- II.A.3.l. “Glass melting furnace” means an emissions unit comprising a refractory vessel in which raw materials are charged, melted at high temperature, refined, and conditioned to produce molten glass.
- II.A.3.m. “Kiln” means the equipment used to remove combined (chemically bound) water and/or gases from mineral material through direct or indirect heating.
- II.A.3.n. “Lightweight aggregate” means the expanded, porous product from heating shales, clays, slates, slags, or other natural materials in a kiln.
- II.A.3.o. “Liquid fuel” means any fuel which is a liquid at standard conditions including but not limited to distillate oils, kerosene and jet fuel. Liquefied gaseous fuels are not liquid fuels.
- II.A.3.p. “Process heater” means an enclosed device using controlled flame and a primary purpose to transfer heat indirectly to a process material or to a heat transfer material for use in a process unit, instead of generating steam. Process heaters are devices in which the combustion gases do not come into direct contact with process materials.
- II.A.3.q. “Reciprocating internal combustion engine” means any reciprocating internal combustion engine which uses reciprocating motion to convert heat energy into mechanical work and which is not used to propel a motor vehicle or a vehicle used solely for competition.

II.A.3.r. “Stationary combustion equipment” means an emissions unit that combusts solid, liquid, or gaseous fuel, generally for the purposes of producing electricity, generating steam, or providing useful heat or energy for industrial, commercial, or institutional use. Stationary combustion equipment includes, but is not limited to, boilers, duct burners, engines, glass melting furnaces, kilns, process heaters, stationary combustion turbines, dryers, furnaces, and ceramic kilns.

II.A.3.s. “Stationary combustion turbine” means a non-mobile, enclosed fossil or other fuel-fired device that is comprised of a compressor, a combustor and a turbine, and in which the flue gas resulting from the combustion of fuel in the combustor passes through the turbine, rotating the turbine. Stationary combustion turbines can be simple cycle or combined cycle and they may or may not include a duct burner.

II.A.4. Emission limitations.

By October 1, 2021, or the applicable date in Section II.A.4.g. for process heaters, no owner or operator of stationary combustion equipment specified in Section II.A.1.a. may cause, allow, or permit NO_x to be emitted in excess of the following emission limitations. When demonstrating compliance using continuous emissions monitoring pursuant to Section II.A.5.b.(i), the following emission limitations are on a 30-day rolling average basis, unless otherwise specified.

By July 20, 2021, or the applicable date in Section II.A.4.g. for process heaters, no owner or operator of stationary combustion equipment specified in Section II.A.1.b. may cause, allow, or permit NO_x to be emitted in excess of the following emission limitations. When demonstrating compliance using continuous emissions monitoring pursuant to Section II.A.5.b.(i), the following emission limitations are on a 30-day rolling average basis, unless otherwise specified.

By May 1, 2022, or May, 1 2023, as specified in Sections II.A.4.g.(i) or II.A.4.g.(ii), no owner or operator of process heaters specified in Section II.A.1.c. may cause, allow, or permit NO_x to be emitted in excess of the following emission limitations. Compliance with the applicable emission limitations contained in Section II.A.4. must be determined according to the applicable methods contained in Sections II.A.5. When demonstrating compliance using continuous emissions monitoring pursuant to Section II.A.5.b.(i), the following emission limitations are on a 30-day rolling average basis, unless otherwise specified.

By May 1, 2024, or the applicable date in Section II.A.4.g. for process heaters, no owner or operator of stationary combustion equipment specified in Sections II.A.1.d. or II.A.1.e. may cause, allow, or permit NO_x to be emitted in excess of the following emission limitations. When demonstrating compliance using continuous emissions monitoring pursuant to Section II.A.5.b.(i), the following emission limitations are on a 30-day rolling average basis, unless otherwise specified.

By May 1, 2025, owners or operators of process heaters subject to Sections II.A.4.g.(iv) and II.A.4.g.(v) must determine compliance with applicable emissions limitations according to Section II.A.5. and, specific to Section II.A.5.b.(i), must determine compliance on a 30-day rolling average basis unless otherwise specified.

By May 1, 2026, no owner or operator of stationary combustion equipment specified in Section II.A.1.f. may cause, allow, or permit NO_x to be emitted in excess of the following emission limitations. When demonstrating compliance using continuous emissions monitoring pursuant to Section II.A.5.b.(i), the following emission limitations are on a 30-day rolling average basis, unless otherwise specified.

II.A.4.a. Boilers.

II.A.4.a.(i) For a gaseous fuel-fired boiler with a maximum design heat input capacity equal to or greater than 100 MMBtu/hr, 0.2 lb/MMBtu of heat input or less than 165 parts per million dry volume corrected to 3% oxygen.

II.A.4.a.(ii) For a liquid fuel-fired boiler with a maximum design heat input capacity equal to or greater than 100 MMBtu/hr, 0.2 lb/MMBtu of heat input or less than 165 parts per million dry volume corrected to 3% oxygen.

II.A.4.a.(iii) For a liquid or gaseous fuel-fired boiler at a major source of NO_x (greater than or equal to 50 tpy NO_x as of January 27, 2020) with a maximum design heat input capacity equal to or greater than 100 MMBtu/hr, 0.2 lb/MMBtu of heat input or less than 165 parts per million dry volume corrected to 3% oxygen.

II.A.4.a.(iv) For a liquid or gaseous fuel-fired boiler at a major source of NO_x (greater than or equal to 50 tpy NO_x as of January 27, 2020) with a maximum design heat input capacity equal to or greater than 50 MMBtu/hr but less than 100 MMBtu/hr, 0.1 lb/MMBtu of heat input or less than 83 parts per million dry volume corrected to 3% oxygen.

II.A.4.a.(v) For a liquid or gaseous fuel-fired boiler at a major source of NO_x (greater than or equal to 25 tpy NO_x as of November 7, 2022) with a maximum design heat input capacity equal to or greater than 20 MMBtu/hr but less than 100 MMBtu/hr, 0.1 lb/MMBtu of heat input or less than 83 parts per million dry volume corrected to 3% oxygen.

II.A.4.a.(vi) For wood-fired boilers at a major source of NO_x (greater than or equal to 25 tpy NO_x as of November 7, 2022) with a maximum design heat input capacity equal to or greater than 20 MMBtu/hr but less than 50 MMBtu/hr, 0.49 lb/MMBtu of heat input.

II.A.4.a.(vii) Boilers subject to the categorical limits in Section II.A.4.a.(i) through (vi) or boilers with a maximum design heat input capacity less than 100 MMBtu/hr must comply with the combustion process adjustment requirements contained in Section II.A.6. while burning wood, gaseous fuel, liquid fuel, or any combination thereof, when required by Section II.A.6.a.

II.A.4.b. Stationary combustion turbines.

II.A.4.b.(i) Stationary combustion turbines with a maximum design heat input capacity equal to or greater than 10 MMBtu/hr and which commenced construction on or before February 18, 2005, must comply with the following NO_x emission limits in Table 1.

Table 1 – NOx limits for stationary combustion turbines constructed on or before February 18, 2005		
Combustion turbine type	Combustion turbine heat input at peak load (HHV)	NOx emission standard
Turbine firing natural gas	> 850 MMBtu/h	15 ppm at 15 percent O ₂ or 54 ng/J of useful output (0.43 lb/MWh)
Turbine firing fuels other than natural gas	> 850 MMBtu/h	42 ppm at 15 percent O ₂ or 160 ng/J of useful output (1.3 lb/MWh).
Turbine	≤ 50 MMBtu/h	150 ppm at 15 percent O ₂ or 1,100 ng/J of useful output (8.7 lb/MWh).
Turbine firing natural gas	> 50 MMBtu/h and ≤ 850 MMBtu/h	42 ppm at 15 percent O ₂ or 250 ng/J of useful output (2.0 lb/MWh).
Turbine firing fuels other than natural gas	> 50 MMBtu/h and ≤ 850 MMBtu/h	96 ppm at 15 percent O ₂ or 590 ng/J of useful output (4.7 lb/MWh).
Turbines operating at less than 75 percent of peak load, turbines operating at temperatures less than 0 °F	≤ 30 MW output	150 ppm at 15 percent O ₂ or 1,100 ng/J of useful output (8.7 lb/MWh).
Turbines operating at less than 75 percent of peak load, turbines operating at temperatures less than 0 °F	> 30 MW output	96 ppm at 15 percent O ₂ or 590 ng/J of useful output (4.7 lb/MWh).
Heat recovery units operating independent of the combustion turbine	All sizes	54 ppm at 15 percent O ₂ or 110 ng/J of useful output (0.86 lb/MWh).

II.A.4.b.(i)(A) For units with heat recovery and CEMS, determine compliance on a 30-day rolling average.

II.A.4.b.(i)(B) For simple cycle turbines with CEMS, determine compliance on a 4-hour rolling average.

II.A.4.b.(i)(C) For operating periods during which multiple emissions standards apply, the applicable standard is the average of the applicable standards during each hour. For hours with multiple emissions standards, the applicable limit for that hour is determined based on the condition that corresponded to the highest emissions standard.

II.A.4.b.(i)(D) Emissions exceeding the NO_x emission limits in Section II.A.4.b.(i) at any time, including during times of startup, shutdown, malfunction, fuel switching, tuning, and testing must be reported as specified in Section II.A.8.a.(i).

II.A.4.b.(ii) Stationary combustion turbines with a maximum design heat input capacity equal to or greater than 10 MMBtu/hr and which commenced construction, modification or reconstruction after February 18, 2005, must comply with the applicable NO_x emission limits in 40 CFR Part 60, Subpart KKKK (October 7, 2020).

II.A.4.b.(iii) Stationary combustion turbines subject to the categorical limits in Section II.A.4.b.(i) or (ii) and stationary combustion turbines with a maximum design heat input capacity less than 10 MMBtu/hr must comply with the combustion process adjustment requirements contained in Section II.A.6. while burning gaseous fuel, liquid fuel, or any combination thereof, when required by Section II.A.6.a.

II.A.4.b.(iv) Stationary combustion turbines, air pollution control equipment, and monitoring equipment must be operated in a manner consistent with good air pollution control practices for minimizing emissions at all times.

II.A.4.c. Lightweight aggregate kilns.

II.A.4.c.(i) For lightweight aggregate kilns with a maximum design heat input capacity equal to or greater than 50 MMBtu/hr, 56.6 pounds of NO_x per hour.

II.A.4.d. Glass melting furnaces, beginning May 1, 2023.

II.A.4.d.(i) 1.2 pounds of NO_x per ton of glass pulled, on a 30-production-day rolling average. A calendar day in which no glass is pulled from the furnace is not a furnace production day. The 30-day rolling average is the sum of all valid hourly NO_x mass emissions recorded by the CEMS or CERM_s during the 30-day period divided by the sum of the glass pulled in the same period. When glass is being pulled, NO_x emissions must be measured continuously in accordance with the applicable monitoring requirements of Section II.A.5.

II.A.4.d.(ii) For periods of time when no glass is pulled, NO_x mass emissions must be calculated as follows and included in the annual mass emissions totals for the furnace.

II.A.4.d.(ii)(A) During initial heating of a furnace using portable burners following the original construction or refractory brick replacement or repair project, the portable burner fuel limit is 8 million standard cubic feet of natural gas. The NO_x emissions resulting from the use of portable burners must be calculated using the total quantity of gas combusted by the portable burners and the uncontrolled NO_x emission factor for the burner heat input design found in Table 1.4-1 of AP42.

II.A.4.d.(ii)(B) When no glass is being pulled and the furnace burners are combusting fuel, NO_x emissions must be measured continuously in accordance with the applicable monitoring requirements of Section II.A.5.

II.A.4.d.(iii) At all times, the furnace must be operated in accordance with good air pollution control practices.

II.A.4.e. Compression ignition RICE.

II.A.4.e.(i) For a compression ignition RICE with a maximum design power output equal to or greater than 500 horsepower, 9 grams per brake horsepower-hour.

II.A.4.e.(ii) For compression ignition RICE subject to Sections II.A.1.d. or II.A.1.e., the more stringent of either Section II.A.4.e.(i) or applicable emission limits of a New Source Performance Standard in 40 CFR Part 60 (July 1, 2022).

II.A.4.e.(iii) Compression ignition RICE subject to the emission limit in Section II.A.4.e.(i) and compression ignition RICE with a maximum design power output less than 500 horsepower must comply with the combustion process adjustment requirements contained in Section II.A.6.

II.A.4.f. Landfill gas or biogas gas fired RICE.

II.A.4.f.(i) For landfill gas or biogas fired RICE with a maximum design power output equal to or greater than 500 horsepower, 1.5 grams per brake horsepower-hour.

II.A.4.g. Process heaters

II.A.4.g.(i) Except as specified in Section II.A.4.g.(ii), by May 1, 2022, natural gas-fired process heaters must comply with the following NOx emission limits in Table 2.

Table 2 – NOx limits for process heaters		
Heat input rate (MMBtu/hr)	Primary fuel type	NOx emission limit (lb/MMBtu)
> 5 and < 100	Natural gas	0.05

II.A.4.g.(ii) Process heaters that require a permitting action or facility shut-down to comply with the NOx emission limits in Table 2 must comply by May 31, 2023.

II.A.4.g.(iii) Process heaters subject under Section II.A.1.e. must comply with the NOx emission limits in Table 2 by May 1, 2024.

II.A.4.g.(iv) By May 1, 2025, the following refinery fuel-fired process heaters must be operated using good combustion practices. In the absence of credible evidence to the contrary, good combustion practices are demonstrated by compliance with the applicable operational practices in Table 3 and performance of the applicable combustion process adjustment requirements in Section II.A.6.

Table 3 – operational practices for refinery fuel-fired process heaters		
Process heater	Primary fuel type	Operational practice
<u>001-0003-002</u>	Refinery gas	Operate low NOx burners
<u>001-0003-012</u>	Refinery gas	Operate ultra-low NOx burners
<u>001-0003-096</u>	Refinery gas	Operate ultra-low NOx burners
<u>001-0003-098</u>	Refinery gas	Operate low NOx burners
<u>001-0003-205</u>	Refinery gas	Operate low NOx burners
<u>001-0003-206</u>	Refinery gas	Operate low NOx burners
<u>001-0003-208</u>	Refinery gas	Operate low NOx burners
<u>001-0003-295</u>	Refinery gas	Operate low NOx burners

II.A.4.g.(v) By May 1, 2025, the following refinery fuel-fired process heaters must comply with the applicable NOx emissions limits in Table 4.

Table 4 – NOx limits for refinery fuel-fired process heaters		
Process heater	Primary fuel type	NOx emission limit
<u>001-0003-097</u>	Refinery gas	40 ppmvd @ 0% O2 on a 30-day rolling average
<u>001-0003-137</u>	Refinery gas	40 ppmvd @ 0% O2 on a 30-day rolling average

II.A.5. Compliance demonstration.

II.A.5.a. Compliance date

II.A.5.a.(i) By October 1, 2021, for stationary combustion equipment that existed at a major source of NOx (greater than or equal to 100 tpy NOx) located in the 8-hour ozone control area as of June 3, 2016, except for process heaters specified in Section II.A.4.g., the owner or operator of an affected unit must determine compliance with the applicable emission limitations contained in Section II.A.4. according to the applicable methods contained in this Section II.A.5.

II.A.5.a.(ii) By July 20, 2021, for stationary combustion equipment specified in Section II.A.1.b., except for process heaters specified in Section II.A.4.g., the owner or operator of an affected unit must determine compliance with the applicable emission limitations contained in Section II.A.4. according to the applicable methods contained in Section II.A.5.

II.A.5.a.(iii) By May 1, 2022, for process heaters specified in Section II.A.4.g.(i)., May 31, 2023, for process heaters specified in Section II.A.4.g.(ii), or May 1, 2025, for process heaters specified in Sections II.A.4.g.(iv) or II.A.4.g.(v), the owner or operator of an affected unit must determine compliance with the applicable emission limitations contained in Section II.A.4. according to the applicable methods contained in Section II.A.5.

II.A.5.a.(iv) By May 1, 2024, for stationary combustion equipment specified in Section II.A.1.d., except for process heaters specified in Section II.A.4.g., the owner or operator of an affected unit must determine compliance with the applicable emission limitations contained in Section II.A.4. according to the applicable methods contained in Section II.A.5.

II.A.5.a.(v) By May 1, 2024, for stationary combustion equipment specified in Section II.A.1.e., except for process heaters specified in Section II.A.4.g., the owner or operator of an affected unit must determine compliance with the applicable emission limitations contained in Section II.A.4. according to the applicable methods contained in Section II.A.5.

II.A.5.a.(vi) By May 1, 2026, for stationary combustion equipment specified in Section II.A.1.f. the owner or operator of an affected unit must determine compliance with the applicable emission limitations contained in Section II.A.4. according to the applicable methods contained in Section II.A.5.

II.A.5.b. Emissions monitoring requirements for major source RACT limits

II.A.5.b.(i) Continuous emission monitoring

II.A.5.b.(i)(A) Owners or operators of an affected unit subject to a NO_x emission limit in Sections II.A.4.a.(i) through II.A.4.a.(iii), II.A.4.c., II.A.4.d., or II.A.4.g.(v) must install, operate and maintain a NO_x CEMS or CERMS to monitor compliance with the applicable emission limit in accordance with this Section II.A.5.b.(i). Owners or operators of affected units' subject to a NO_x emission limit in Sections II.A.4.b., II.A.4.e., or II.A.4.g. may install, operate and maintain a NO_x CEMS or CERMS to monitor compliance with the applicable emission limit in accordance with this Section II.A.5.b.(i) in lieu of performance testing pursuant to Section II.A.5.b.(ii).

II.A.5.b.(i)(A)(1) The owner or operator of an affected unit that is subject to or becomes subject to the monitoring requirements of 40 CFR part 75 and 40 CFR part 75, Appendices A to I (July 19, 2018), must use those monitoring methods and specifications for monitoring NO_x emissions for purposes of this Section II.A.5. and for demonstrating compliance with Section II.A.4. The missing data substitution procedures and bias adjustment requirements of 40 CFR Part 75 (July 19, 2018) do not apply for purposes of demonstrating compliance with Section II.A.4. or this Section II.A.5.

II.A.5.b.(i)(A)(2) For an affected unit equipped with a NO_x CEMS or CERMS for purposes of demonstrating compliance with an applicable subpart of 40 CFR Part 60 (July 1, 2022), the owner or operator must use the definition of operating day, data averaging methodology, and data validation requirements of the applicable subpart of 40 CFR Part 60 for purposes of demonstrating compliance with an applicable emission limit in Section II.A.4. The owner or operator must calibrate, maintain, and operate the CEMS or CERMS and validate emissions data according to the applicable requirements of 40 CFR Part 60, Section 60.13 (July 1, 2022), the performance specifications in 40 CFR Part 60, Appendix B (October 7, 2020), and the quality assurance procedures of 40 CFR Part 60, Appendix F (October 7, 2020).

II.A.5.b.(i)(A)(3) For an affected unit that is not equipped with a NO_x CEMS or CERMS for purposes of demonstrating compliance with 40 CFR Part 60 (July 1, 2022) or Part 75 (July 19, 2018), the owner or operator must install, operate, and maintain a NO_x CEMS or CERMS that measures emissions in terms of the applicable emission limitation and must calibrate, maintain, and operate the CEMS or CERMS and validate emissions data according to the applicable provisions of 40 CFR Part 60, Section 60.13 (July 19, 2018), the performance specifications in 40 CFR Part 60, Appendix B (October 7, 2020), and the quality assurance procedures of 40 CFR Part 60, Appendix F (October 7, 2020). The owner or operator must use the following methodology for purposes of demonstrating compliance with an applicable 30-day rolling average emission limit in Section II.A.4.

II.A.5.b.(i)(A)(3)(a) A unit operating day is a calendar day when any fuel is combusted in the affected unit.

II.A.5.b.(i)(A)(3)(b) 30-day rolling average emission rates must be calculated as the arithmetic average emissions rates determined by the CEMS or CERMS for all hours the affected unit combusted any fuel from the current unit operating day and the prior 29 unit operating days.

II.A.5.b.(i)(A)(4) When an affected unit utilizes a common flue gas stack system with one or more affected units, but no non-affected units, the owner or operator must follow the applicable procedures of 40 CFR Part 75, Appendix F (July 19, 2018) for the determination of all sampling locations and apportionment of emissions to an individual affected unit.

II.A.5.b.(i)(B) Owners or operators of a stationary combustion turbine subject to a NO_x emission limit in Section II.A.4.b. must comply with

II.A.5.b.(i)(B)(1) The applicable monitoring requirements in 40 CFR Part 60, Subpart GG (October 7, 2020) for turbines which commenced construction on or before February 18, 2005.

II.A.5.b.(i)(B)(2) The applicable monitoring requirements in 40 CFR Part 60, Subpart KKKK (October 7, 2020) for turbines which commenced construction after February 18, 2005.

II.A.5.b.(ii) Initial and periodic performance testing

II.A.5.b.(ii)(A) An owner or operator of a stationary combustion turbine subject to 40 CFR Part 60, Subparts GG or KKKK (October 7, 2020) that has used and continues to use performance testing to demonstrate compliance with either Subpart GG or KKKK (October 7, 2020), as applicable, may use those performance testing procedures to demonstrate continued compliance with an applicable limitation contained in Section II.A.4.b., thereby satisfying the requirements of this Section II.A.5.b.(ii).

II.A.5.b.(ii)(B) An owner or operator of a process heater subject to a NO_x emission limit in Section II.A.4.g. must

II.A.5.b.(ii)(B)(1) For natural gas-fired heaters greater than or equal to 100 MMBtu/hr, conduct an initial performance test in accordance with Sections II.A.5.b.(ii)(D)(1), II.A.5.b.(ii)(D)(4), and II.A.5.b.(ii)(E) by April 1, 2022, or by November 30, 2023, for process heaters specified in Section II.A.4.g.(ii), and conduct subsequent performance tests in accordance with Sections II.A.5.b.(ii)(D)(1), II.A.5.b.(ii)(D)(4), and II.A.5.b.(ii)(E) every 2 years thereafter.

II.A.5.b.(ii)(B)(2) For natural gas-fired process heaters greater than or equal to 50 MMBtu/hr and less than 100 MMBtu/hr, conduct an initial performance test in accordance with Sections II.A.5.b.(ii)(D)(1), II.A.5.b.(ii)(D)(4), and II.A.5.b.(ii)(E) by April 1, 2022, or by November 30, 2023, for process heaters specified in Section II.A.4.g.(ii), and comply with the combustion process adjustment requirements in Section II.A.6. thereafter.

II.A.5.b.(ii)(B)(3) For natural gas-fired process heaters greater than or equal to 5 MMBtu/hr and less than 50 MMBtu/hr and refinery gas-fired process heaters greater than or equal to 5 MMBtu/hr and less than 100 MMBtu/hr, comply with the combustion process adjustment requirements in Section II.A.6.

II.A.5.b.(ii)(B)(4) For a group of process heaters that vent to a common stack, the owner or operator may either assess compliance for the heaters individually by performing a separate emission test of each heater in the duct leading from the heater to the common stack or may perform a single emission test in the common stack. The owner or operator must include in the test protocol for these units a definition of representative conditions for performance testing purposes.

II.A.5.b.(ii)(B)(5) For natural gas-fired process heaters subject under Sections II.A.1.d. or II.A.1.e. and greater than or equal to 50 MMBtu/hr and less than 100 MMBtu/hr, conduct an initial performance test in accordance with Sections II.A.5.b.(ii)(D)(1), II.A.5.b.(ii)(D)(4), and II.A.5.b.(ii)(E) by April 1, 2024, and comply with the combustion process adjustment requirements in Section II.A.6. thereafter.

II.A.5.b.(ii)(B)(6) Performance tests conducted in accordance with Sections II.A.5.b.(ii)(D)(1) through II.A.5.b.(ii)(D)(3) and II.A.5.b.(ii)(E) within three (3) years of the applicable performance testing date in Sections II.A.5.b.(ii)(B)(1) or II.A.5.b.(ii)(B)(2) will satisfy the initial performance testing requirement.

II.A.5.b.(ii)(B)(7) As an alternative to the requirements in Sections II.A.5.b.(ii)(B)(1), II.A.4.b.(ii)(B)(2), II.A.5.b.(ii)(B)(4), and II.A.5.b.(ii)(B)(5), the owner or operator may install, operate, and maintain a NO_x CEMS or CERMS in accordance with Sections II.A.5.b(i)(A)(1) through II.A.5.b.(i)(A)(4) to monitor compliance with the applicable emission limit.

II.A.5.b.(ii)(C) An owner or operator of a boiler subject to a NO_x limit under Sections II.A.4.a.(v) or II.A.4.a.(vi), conduct an initial performance test in accordance with Sections II.A.5.b.(ii)(D)(1), II.A.5.b.(ii)(D)(4), and II.A.5.b.(ii)(E) by April 1, 2024, and comply with the combustion process adjustment requirements in Section II.A.6. thereafter.

II.A.5.b.(ii)(D) Except as otherwise provided for in Sections II.A.5.b.(ii)(A), II.A.5.b.(ii)(B), or II.A.5.b.(ii)(C), the owner or operator of an affected unit subject to a NO_x emission limitation contained in Sections II.A.4.a.(iv), II.A.4.b., or II.A.4.e. that is not equipped with NO_x CEMS or CERMS, must conduct an initial performance test and subsequent performance tests every 2 years thereafter, according to the following requirements, as applicable, to determine the affected unit's NO_x emission rate for each fuel fired in the affected unit.

II.A.5.b.(ii)(D)(1) A performance test is not required for a fuel used only for startup or for a fuel constituting less than 2% of the unit's annual heat input, as determined at the end of each calendar year.

II.A.5.b.(ii)(D)(2) Initial performance test must include a determination of the capacity load point of the unit's maximum NO_x emissions rate based on one 30-minute test run at each capacity load point for which the unit is operated, other than for startup and shutdown, in the load ranges of 20 to 30%, 45 to 55%, and 70 to 100%. Subsequent performance tests must be performed within the capacity load range determined to result in the maximum NO_x emissions rate.

II.A.5.b.(ii)(D)(3) Performance tests must determine compliance with Section II.A.4. based on the average of three 60-minute test runs performed at the capacity load determined in Section II.A.5.b.(ii)(D)(2).

II.A.5.b.(ii)(D)(4) Initial performance test must be conducted at both high and low load capacity. If site operations do not allow testing at high and low loads, the initial performance test must be conducted at the highest achievable load that site conditions allow. The owner or operator must submit a summary of six months of operating performance with the test protocol supporting the testing load(s). Subsequent performance tests must be performed within the capacity load range determined to result in the maximum NO_x emissions rate. Performance tests must determine compliance based on the average of three 60-minute test runs.

II.A.5.b.(ii)(E) The owner or operator of an affected unit subject to a NO_x emission limitation contained in Section II.A.4.f. that is not equipped with NO_x CEMS or CERMS, must conduct an initial performance test by May 1, 2025, and subsequent performance tests every 3 years thereafter, according to the requirements in Sections II.A.5.b.(ii)(D)(1) through II.A.5.b.(ii)(D)(4), as applicable, to determine the affected unit's NO_x emission rate for each fuel fired in the affected unit. In lieu of subsequent triennial performance tests, the owner or operator may conduct semi-annual portable analyzer monitoring for NO_x conducted using the Division's Portable Analyzer Monitoring Protocol (March 2006). A performance test conducted on the engine in accordance with 40 C.F.R. Part 60 (July 1, 2023) between January 1, 2022, and April 30, 2025, will satisfy the initial performance test requirement.

II.A.5.b.(ii)(F) All performance tests must be conducted in accordance with EPA test methods and a test protocol submitted to the Division for review at least thirty (30) days prior to testing and in accordance with AQCC Common Provisions Regulation Section II.C.

II.A.5.b.(iii) For affected units' subject to a production-based or output based emission limit contained in Section II.A.4. (e.g. lb of NO_x/ton of product), the owner or operator must install, operate, and maintain monitoring equipment for measuring and recording the affected unit's production or output, on an hourly basis, in units consistent with the applicable emission limitation.

II.A.5.b.(iv) Where measuring fuel use is necessary to calculate an emission rate in the units of the applicable standard, fuel flowmeters must be installed, calibrated, maintained, and operated according to manufacturer's instructions for measuring and recording heat input in terms of the applicable emission limitation. Alternatively, fuel flowmeters that meet the installation, certification, and quality assurance requirements of 40 CFR Part 75, Appendix D (July 19, 2018) are acceptable for demonstrating compliance with this section. The installation of fuel-flowmeters is not required where emissions of NO_x in terms of the applicable standard can be calculated in accordance with applicable provisions of EPA Method 19 (March 29, 2023) or where the standard is concentration based (e.g. parts per million dry volume corrected for oxygen).

II.A.6. Combustion process adjustment

II.A.6.a. Applicability

II.A.6.a.(i) As of January 1, 2017, this Section II.A.6. applies to boilers, duct burners, process heaters, stationary combustion turbines, and stationary reciprocating internal combustion engines with uncontrolled actual emissions of NO_x equal to or greater than five (5) tons per year that existed at major sources of NO_x (greater than or equal to 100 tpy NO_x) located in the 8-hour ozone control area as of June 3, 2016.

II.A.6.a.(ii) As of May 1, 2020, this Section II.A.6. applies to boilers, duct burners, process heaters, stationary combustion turbines, stationary reciprocating internal combustion engines, dryers, furnaces, and ceramic kilns with uncontrolled actual emissions of NO_x equal to or greater than five (5) tons per year that existed at major sources of NO_x (greater than or equal to 50 tpy NO_x) located in the 8-hour ozone control area as of January 27, 2020, that is not already subject as provided under Section II.A.6.a.(i).

- II.A.6.a.(iii) As of May 1, 2022, or May 31, 2023, for process heaters specified in Section II.A.4.g.(ii), this Section II.A.6. applies to process heaters with uncontrolled actual emissions of NO_x equal to or greater than five (5) tons per year that existed at sources that emit, or have the potential to emit, NO_x emissions greater than or equal to 25 tpy NO_x located in the 8-hour ozone control area as of July 20, 2021, that is not already subject as provided under Sections II.A.6.a.(i) or II.A.6.a.(ii).
- II.A.6.a.(iv) As of February 14, 2023, this Section II.A.6. applies to boilers, duct burners, stationary combustion turbines, stationary reciprocating internal combustion engines, dryers, furnaces, ceramic kilns, and process heaters with uncontrolled actual emissions of NO_x equal to or greater than five (5) tons per year that existed at major sources of NO_x (greater than or equal to 25 tpy NO_x) located in the 8-hour ozone control area as of November 7, 2022, that is not already subject as provided under Sections II.A.6.a.(i) through II.A.6.a.(iii).
- II.A.6.a.(v) As of February 14, 2023, this Section II.A.6. applies to boilers, duct burners, stationary combustion turbines, stationary reciprocating internal combustion engines, dryers, furnaces, ceramic kilns, and process heaters with uncontrolled actual emissions of NO_x equal to or greater than five (5) tons per year that existed at major sources of NO_x (greater than or equal to 100 tpy NO_x) in northern Weld County as of November 7, 2022.
- II.A.6.a.(vi) As of May 1, 2026, this Section II.A.6. applies to boilers, duct burners, stationary combustion turbines, stationary reciprocating internal combustion engines, dryers, furnaces, ceramic kilns, and process heaters with uncontrolled actual emissions of NO_x equal to or greater than five (5) tons per year that existed at major sources of NO_x (greater than or equal to 50 tpy NO_x) in northern Weld County as of July 24, 2024.

II.A.6.b. Combustion process adjustment

- II.A.6.b.(i) When burning the fuel that provides the majority of the heat input since the last combustion process adjustment and when operating at a firing rate typical of normal operation, the owner or operator must conduct the following inspections and adjustments of boilers and process heaters, as applicable:
- II.A.6.b.(i)(A) Inspect the burner and combustion controls and clean or replace components as necessary.
 - II.A.6.b.(i)(B) Inspect the flame pattern and adjust the burner or combustion controls as necessary to optimize the flame pattern.
 - II.A.6.b.(i)(C) Inspect the system controlling the air-to-fuel ratio and ensure that it is correctly calibrated and functioning properly.
 - II.A.6.b.(i)(D) Measure the concentration in the effluent stream of carbon monoxide and nitrogen oxide in ppm, by volume, before and after the adjustments in Sections II.A.6.b.(i)(A) through (C). Measurements may be taken using a portable analyzer.
- II.A.6.b.(ii) The owner or operator of a duct burner must inspect duct burner elements, baffles, support structures, and liners and clean, repair, or replace components as necessary.
- II.A.6.b.(iii) The owner or operator of a stationary combustion turbine must conduct the following inspections and adjustments, as applicable:
- II.A.6.b.(iii)(A) Inspect turbine inlet systems and align, repair, or replace components as necessary.
 - II.A.6.b.(iii)(B) Inspect the combustion chamber components, combustion liners, transition pieces, and fuel nozzle assemblies and clean, repair, or replace components as necessary.
 - II.A.6.b.(iii)(C) When burning the fuel that provides the majority of the heat input since the last combustion process adjustment and when operating at a firing rate typical of normal operation, confirm proper setting and calibration of the combustion controls.

- II.A.6.b.(iv) The owner or operator of a stationary internal combustion engine must conduct the following inspections and adjustments, as applicable:
 - II.A.6.b.(iv)(A) Change oil and filters as necessary.
 - II.A.6.b.(iv)(B) Inspect air cleaners, fuel filters, hoses, and belts and clean or replace as necessary.
 - II.A.6.b.(iv)(C) Inspect spark plugs and replace as necessary.
- II.A.6.b.(v) The owner or operator of a dryer or furnace must inspect the burner and combustion controls and adjust, clean, and/or replace components as necessary.
- II.A.6.b.(vi) The owner or operator of a ceramic kiln must inspect and maintain the combustion controls and adjust the burners as necessary to ensure a proper air-to-fuel ratio. At units where entry into a piece of process equipment is required to complete the combustion process adjustment, in-kiln inspections and adjustments are required only during planned entries.
- II.A.6.b.(vii) The owner or operator must operate and maintain the boiler, duct burner, process heater, stationary combustion turbine, stationary internal combustion engine, dryer, furnace, or ceramic kiln consistent with manufacturer's specifications, if available, or good engineering and maintenance practices.
- II.A.6.b.(viii) Frequency

II.A.6.b.(viii)(A) The owner or operator of boilers, duct burners, process heaters, stationary combustion turbines, and stationary reciprocating internal combustion engines with uncontrolled actual emissions of NO_x equal to or greater than five (5) tons per year that existed at major sources of NO_x (greater than or equal to 100 tpy NO_x) as of June 3, 2016, must conduct the initial combustion process adjustment by April 1, 2017. An owner or operator may rely on a combustion process adjustment conducted in accordance with applicable requirements and schedule of a New Source Performance Standard in 40 CFR Part 60 (November 17, 2016) or National Emission Standard for Hazardous Air Pollutants in 40 CFR Part 63 (November 17, 2016) to satisfy the requirement to conduct an initial combustion process adjustment by April 1, 2017.

II.A.6.b.(viii)(B) The owner or operator of boilers, duct burners, process heaters, stationary combustion turbines, stationary reciprocating internal combustion engines, dryers, furnaces, and ceramic kilns with uncontrolled actual emissions of NO_x equal to or greater than five (5) tons per year that existed at major sources of NO_x (greater than or equal to 50 tpy NO_x) as of January 27, 2020, must conduct the initial combustion process adjustment by May 1, 2020. An owner or operator may rely on a combustion process adjustment conducted in accordance with applicable requirements and schedule of a New Source Performance Standard in 40 CFR Part 60 (December 19, 2019) or National Emission Standard for Hazardous Air Pollutants in 40 CFR Part 63 (December 19, 2019) to satisfy the requirement to conduct an initial combustion process adjustment by May 1, 2020.

II.A.6.b.(viii)(C) The owner or operator of process heaters with uncontrolled actual emissions that emit, or have the potential to emit, NO_x equal to or greater than five (5) tons per year that existed at sources of NO_x located in the 8-hour ozone control area emissions greater than or equal to 25 tpy NO_x as of July 20, 2021, must conduct an initial combustion process adjustment by January 1, 2022, or January 1, 2024, for process heaters specified in Section II.A.4.g.(ii). An owner or operator may rely on a combustion process adjustment conducted in accordance with applicable requirements and schedule of a New Source Performance Standard in 40 CFR Part 60 (July 1, 2021) or National Emission Standard for Hazardous Air Pollutants in 40 CFR Part 63 (July 1, 2021) to satisfy the requirement to conduct an initial combustion process adjustment by January 1, 2022.

II.A.6.b.(viii)(D) The owner or operator of boilers, duct burners, stationary combustion turbines, stationary reciprocating internal combustion engines, dryers, furnaces, and ceramic kilns with uncontrolled actual emissions of NO_x equal to or greater than five (5) tons per year that existed at major sources of NO_x (greater than or equal to 25 tpy NO_x) located in the 8-hour ozone control area as of November 7, 2022, must conduct the initial combustion process adjustment by May 1, 2024, unless a performance test is required under Section II.A.5.b.(ii) within one year after the initial performance test. An owner or operator may rely on a combustion process adjustment conducted in accordance with applicable requirements and schedule of a New Source Performance Standard in 40 CFR Part 60 (July 1, 2022) or National Emission Standard for Hazardous Air Pollutants in 40 CFR Part 63 (July 1, 2022) to satisfy the requirement to conduct an initial combustion process adjustment.

II.A.6.b.(viii)(E) The owner or operator of boilers, duct burners, stationary combustion turbines, stationary reciprocating internal combustion engines, dryers, furnaces, ceramic kilns, and process heaters with uncontrolled actual emissions of NO_x equal to or greater than five (5) tons per year that existed at major sources of NO_x (greater than or equal to 100 tpy NO_x) in northern Weld County as of November 7, 2022, must conduct the initial combustion process adjustment by May 1, 2024, unless a performance test is required under Section II.A.5.b.(ii) within one year after the initial performance test. An owner or operator may rely on a combustion process adjustment conducted in accordance with applicable requirements and schedule of a New Source Performance Standard in 40 CFR Part 60 (July 1, 2022) or National Emission Standard for Hazardous Air Pollutants in 40 CFR Part 63 (July 1, 2022) to satisfy the requirement to conduct an initial combustion process adjustment.

II.A.6.b.(viii)(F) The owner or operator of boilers, duct burners, stationary combustion turbines, stationary reciprocating internal combustion engines, dryers, furnaces, ceramic kilns, and process heaters with uncontrolled actual emissions of NO_x equal to or greater than five (5) tons per year that existed at major sources of NO_x (greater than or equal to 50 tpy NO_x) in northern Weld County as of July 24, 2024, must conduct the initial combustion process adjustment by May 1, 2026, unless a performance test is required under Section II.A.5.b.(ii) within one year after the initial performance test. An owner or operator may rely on a combustion process adjustment conducted in accordance with applicable requirements and schedule of a New Source Performance Standard in 40 CFR Part 60 (July 1, 2025) or National Emission Standard for Hazardous Air Pollutants in 40 CFR Part 63 (July 1, 2025) to satisfy the requirement to conduct an initial combustion process adjustment.

II.A.6.b.(viii)(G) The owner or operator must conduct subsequent combustion process adjustments at least once every twelve (12) months after the initial combustion adjustment, or on the applicable schedule according to Sections II.A.6.c.(1). or II.A.6.c.(ii).

II.A.6.b.(viii)(H) Beginning January 1, 2022, the owner or operator of process heaters at a refinery must conduct subsequent combustion process adjustments at least once every six (6) months after the initial combustion adjustment, or on the applicable schedule according to Sections II.A.6.c.(i). or II.A.6.c.(ii).

II.A.6.c. As an alternative to the requirements described in Sections II.A.6.b.(i) through II.A.6.b.(viii):

II.A.6.c.(i) The owner or operator may conduct the combustion process adjustment according to the manufacturer recommended procedures and schedule; or

II.A.6.c.(ii) The owner or operator of combustion equipment that is subject to and required to conduct a periodic tune-up or combustion adjustment by the applicable requirements of a New Source Performance Standard in 40 CFR Part 60 (July 1, 2025) or National Emission Standard for Hazardous Air Pollutants in 40 CFR Part 63 (July 1, 2025) may conduct tune-ups or adjustments according to the schedule and procedures of the applicable requirements of 40 CFR Part 60 (July 1, 2025) or 40 CFR Part 63 (July 1, 2025).

II.A.7. Recordkeeping. The following records must be kept for a period of five years and made available to the Division upon request:

II.A.7.a. The applicable emission limit and calculated heat input weighted emission limit for stationary combustion equipment demonstrating compliance for multiple fuels.

II.A.7.b. The 30-day rolling average emission rate calculated on a daily basis for sources using CERMS to comply with Section II.A.

II.A.7.c. The type and amount of fuel used.

II.A.7.d. The stationary combustion equipment's annual capacity factor on a calendar year basis.

- II.A.7.e. All records generated to comply with the reporting requirements contained in Section II.A.8.
- II.A.7.f. For stationary combustion equipment subject to the combustion process adjustment requirements in Section II.A.6., the following recordkeeping requirements apply:
 - II.A.7.f.(i) The owner or operator must create a record once every calendar year identifying the combustion equipment at the source subject to Section II.A. and including for each combustion equipment:
 - II.A.7.f.(i)(A) The date of the adjustment;
 - II.A.7.f.(i)(B) Whether the combustion process adjustment under Sections II.A.6.b.(i) through II.A.6.b.(vi) was followed, and what procedures were performed;
 - II.A.7.f.(i)(C) Whether a combustion process adjustment under Sections II.A.6.c.(i). and II.A.6.c.(ii). was followed, what procedures were performed, and what New Source Performance or National Emission Standard for Hazardous Air Pollutants applied, if any; and
 - II.A.7.f.(i)(D) A description of any corrective action taken.
 - II.A.7.f.(i)(E) If the owner or operator conducts the combustion process adjustment according to the manufacturer recommended procedures and schedule and the manufacturer specifies a combustion process adjustment on an operation time schedule, the hours of operation.
 - II.A.7.f.(i)(F) If multiple fuels are used, the type of fuel burned and heat input provided by each fuel.
 - II.A.7.f.(ii) The owner or operator must retain manufacturer recommended procedures, specifications, and maintenance schedule if utilized under Section II.A.6.c.(i). for the life of the equipment.

II.A.7.f.(iii) As an alternative to the requirements described in Section II.A.7.f.(i), the owner or operator may comply with applicable recordkeeping requirements related to combustion process adjustments conducted according to a New Source Performance Standard in 40 CFR Part 60 (July 1, 2025) or National Emission Standard for Hazardous Air Pollutants in 40 CFR Part 63 (July 1, 2025).

II.A.7.g. All sources qualifying for an exemption under Section II.A.2. must maintain all records necessary to demonstrate that an exemption applies.

II.A.7.h. Records of semi-annual portable analyzer monitoring, including the date, engine settings on the date of the monitoring, and documentation of the results of the monitoring.

II.A.8. Reporting

II.A.8.a. For affected units demonstrating compliance with Section II.A.4. using CEMS or CERMS in accordance with Section II.A.5.b.(i)(A), the owner or operator must submit to the Division the following information:

II.A.8.a.(i) Quarterly or semi-annual excess emissions reports no later than the 30th day following the end of each semi-annual or quarterly period, as applicable. Excess emissions means emissions that exceed the applicable limitations contained in Section II.A.4. Excess emission reports must include the information specified in 40 CFR Part 60, Section 60.7(c) (July 1, 2018).

II.A.8.b. For affected units demonstrating compliance with Section II.A.4 using performance testing pursuant to Section II.A.5.b.(ii), the owner or operator must submit to the Division the following information:

II.A.8.b.(i) Performance test or portable analyzer testing reports within 60 days after completion of the performance test program or portable analyzer testing. All performance test reports must determine compliance with the applicable emission limitations set by Section II.A.4.

- II.A.8.c. Beginning in 2025, the owner or operator of combustion equipment subject to combustion process adjustment requirements in Section II.A.6. but not Sections II.A.4.a.(i) through II.A.4.a.(vi), II.A.4.b.(i), II.A.4.b.(ii), II.A.4.c.(i), II.A.4.d., II.A.4.e.(i), II.A.4.e.(ii), II.A.4.f., II.A.4.g.(i), or II.A.4.g.(v) must submit with the semi-annual report required by the facility's operating permit documentation of
- II.A.8.d. Beginning in 2026, the owner or operator of combustion equipment identified in Section II.A.1.f. and subject to combustion process adjustment requirements in Section II.A.6. but not Sections II.A.4.a.(i) through II.A.4.a.(vi), II.A.4.b.(i), II.A.4.b.(ii), II.A.4.c.(i), II.A.4.d., II.A.4.e.(i), II.A.4.e.(ii), II.A.4.f., II.A.4.g.(i), or II.A.4.g.(v) must submit with the semi-annual report required by the facility's operating permit documentation of compliance with the combustion process adjustment requirements.

III. Control of Emissions from Specific Major Sources of VOC and/or NO_x in the 8-hour Ozone Control Area

III.A. Specific major sources of VOC and/or NO_x (greater than or equal to 100 tpy) as of June 3, 2016, located in the 8-hour Ozone Control Area.

III.A.1. Stationary internal combustion engines at the following major sources must comply with applicable NO_x emission limits and associated monitoring, recordkeeping, and reporting requirements in 40 CFR Part 60, Subpart IIII (July 1, 2016), 40 CFR Part 60, Subpart JJJJ (July 1, 2016), and/or 40 CFR Part 63, Subpart ZZZZ (July 1, 2016) as expeditiously as practicable, but no later than January 1, 2017:

- III.A.1.a. National Reconnaissance Office (NRO) – Aerospace Data Facility (005-0028) – engines (pt 128, 139, 144).
- III.A.1.b. Colorado State University (069-0011) – engines (pt 024, 035, 036, 037, 038, 040, 043, 052).
- III.A.1.c. DCP Midstream, Greeley (123-0099) – engine (pt 102).
- III.A.1.d. DCP Midstream, Kersey/Mewbourn (123-0090) – engine (pt 101).
- III.A.1.e. DCP Midstream, Spindle (123-0015) – engines (pt 059, 075).
- III.A.1.f. IBM (013-0006) – engines (pt 092, 094).
- III.A.1.g. Owens-Brockway (123-4406) – engine (pt 024).

- III.A.1.h. Plains End (059-0864) – engine (pt 005).
- III.A.1.i. PSCo Cherokee (001-0001) – engine (pt 031).
- III.A.1.j. Spindle Hill (123-5468) – engine (pt 005).
- III.A.1.k. Suncor (001-0003) – engines (pt 150, 151).
- III.A.1.l. Timberline Energy (123-0079) – engines (pt 010, 011).
- III.A.2. Cemex Construction Materials (013-0003) must comply with applicable THC requirements and associated monitoring, recordkeeping, and reporting in 40 CFR Part 63, Subpart LLL (July 1, 2016) as expeditiously as practicable, but no later than January 1, 2017.
- III.A.3. Denver Regional Landfill and Front Range Landfill (123-0079) (pt 007, 013) must comply with applicable flare requirements in 40 CFR Part 60, Subpart WWW (July 1, 2016) as expeditiously as practicable, but no later than January 1, 2017.
- III.B. Specific major sources of VOC and/or NO_x (greater than or equal to 50 tpy) as of January 27, 2020, located in the 8-hour Ozone Control Area.
 - III.B.1. Stationary internal combustion engines at the following major sources must comply with applicable NO_x emission limits and associated monitoring, recordkeeping, and reporting requirements in 40 CFR Part 60, Subpart IIII (July 1, 2016), 40 CFR Part 60, Subpart JJJJ (July 1, 2016), and/or 40 CFR Part 63, Subpart ZZZZ (January 30, 2013) as expeditiously as practicable, but no later than July 1, 2021:
 - III.B.1.a. University of Colorado Denver, Anschutz Medical Campus (001-0106) – engines (pts 011, 012, 013, 014, 015, 016, 017, 018, 020, 021).
 - III.B.1.b. Centura Health St. Anthony (059-1511) – engines (pts 002, 003).
 - III.B.2. Flares at the following major sources must comply with applicable flare requirements in 40 CFR Part 60, Section 60.18 (December 22, 2008) as expeditiously as practicable, but no later than July 1, 2021.
 - III.B.2.a. Waste Management of Colorado Denver Arapahoe Disposal Site (005-1291) (pt 003).
 - III.B.3. Front Range Energy (123-5097) must comply with applicable monitoring, recordkeeping, and reporting in 40 CFR Part 60, Subpart VV (July 1, 2019) as expeditiously as practicable, but no later than July 1, 2021.

III.C. Specific major sources of VOC and/or NO_x (greater than or equal to 100 tpy) as of November 7, 2022, located in northern Weld County.

III.C.1. Beginning February 14, 2023, stationary combustion turbines with a maximum design heat input capacity equal to or greater than 10 MMBtu/hr located at the Cheyenne Compressor Station and Cheyenne Plains Compressor Station (123-0051).

III.C.1.a. Must comply with the following 1-hour average NO_x emission standards.

III.C.1.a.(i) Except as specified in Section III.C.1.a.(iii), turbines 123-0051-015: 24.5 ppmvd at 15 percent O₂.

III.C.1.a.(ii) Except as specified in Section III.C.1.a.(iii), turbines 123-0051-024 and 123-0051-014: 15 ppmvd at 15 percent O₂.

III.C.1.a.(iii) When temperatures are greater than -20 degrees and less than 0 degrees Fahrenheit: 42 ppmvd at 15 percent O₂. When temperatures are less than -20 degrees Fahrenheit: 120 ppmvd at 15 percent O₂.

III.C.1.b. Must comply with the combustion process adjustment requirements contained in Section II.A.6.b.(iii).

III.C.1.b.(i) The owner or operator must conduct an initial combustion process adjustment by May 1, 2024.

III.C.1.b.(ii) The owner or operator must conduct subsequent combustion process adjustments at least once every twelve (12) months after the initial combustion adjustment.

III.C.1.c. Must be operated in a manner consistent with good air pollution control practices for minimizing emissions at all times.

III.C.1.d. Must comply with the following monitoring requirements.

III.C.1.d.(i) Conduct portable monitoring each calendar quarter using a portable flue gas analyzer.

III.C.1.d.(ii) At least annually, conduct portable monitoring at the temperatures specified in Section III.C.1.a.(iii), unless ambient conditions or extended periods at those temperatures are not sufficient to conduct the monitoring.

- III.C.1.e. The following records must be kept for a period of five (5) years and made available to the Division upon request.
 - III.C.1.e.(i) Records of the twelve-month rolling total of emissions.
 - III.C.1.e.(ii) Records of the number of hours that the turbine operates when the ambient temperature meets the criteria in Section II.C.1.a.(iii).
 - III.C.1.e.(iii) The combustion process adjustment records specified in Section II.A.7.f.
- III.C.2. Glycol dehydrators at the Cheyenne Compressor Station and Cheyenne Plains Compressor Station (123-0051) must comply with applicable control, monitoring, recordkeeping, and reporting in 40 CFR Part 63, Subpart HHH (November 19, 2020) beginning February 14, 2023.
- III.C.3. Flares at the Cheyenne Compressor Station and Cheyenne Plains Compressor Station (123-0051) must comply with applicable flare requirements in 40 CFR Part 63, Section 63.11 (December 22, 2008) beginning February 14, 2023.
- III.C.4. Beginning May 1, 2023, stationary internal combustion engines at the Cheyenne Compressor Station and Cheyenne Plains Compressor Station (123-0051) (pts 001, 007, 008, 011, 012, 013, 017) must comply with the following NOx emission limits, monitoring, and recordkeeping requirements.
 - III.C.4.a. NOx emissions standards.
 - III.C.4.a.(i) 2-stroke lean burn engines: 3.0 g/hp-hr.
 - III.C.4.a.(ii) 4-stroke lean burn engines: 1.2 g/hp-hr.
 - III.C.4.b. Conduct semi-annual portable analyzer monitoring for NOx.
 - III.C.4.c. Comply with the combustion process adjustment requirements in Section I.D.5.e.(iv).
 - III.C.4.d. Recordkeeping. The following records must be kept for a period of five (5) years and made available to the Division upon request.
 - III.C.4.d.(i) Records of semi-annual portable analyzer monitoring, including the date, engine settings on the date of the monitoring, and documentation of the results of the monitoring.

III.C.4.d.(ii) The combustion process adjustment records specified in Section I.D.5.f.(vi).

IV. Control of Emissions from Breweries in the 8-hour Ozone Control Area

IV.A. Requirements for Brewing Operations

IV.A.1. Applicability

Except as provided in Section IV.A.2., the requirements of Section IV. apply to owners or operators of breweries that existed at a major source of VOC (greater than or equal to 100 tpy VOC) as of June 3, 2016, located in the 8-hour Ozone Control Area.

IV.A.2. Exemptions

The following emissions units are exempt from Sections IV.A.4. through IV.A.7. but must be maintained and operated in a manner consistent with good air pollution control practices for minimizing emissions. Owners or operators must also maintain records necessary to demonstrate that an exemption applies and make such records available to the Division upon request.

Once an emissions unit at a brewery no longer qualifies for an exemption, the owner or operator must comply with the applicable requirements of Sections IV.A.4. through IV.A.7. as expeditiously as practicable but no later than twelve (12) months after the exemption no longer applies, except as specified in Sections IV.A.2.c. and IV.A.2.d.

IV.A.2.a. An emissions unit subject to a work practice or emission control requirement in another federally enforceable section of Regulation Number 7, Number 24, Number 25, and Number 26.

IV.A.2.b. An emissions unit with total uncontrolled actual emissions less than two (2) tons per year VOC on a calendar year basis.

IV.A.2.c. Equipment or activities related to research and development. Research and development ends when the product is sold or offered for sale.

IV.A.2.d. Newly installed, upgraded, or replaced packaging operations for a duration of six months after startup.

IV.A.3. Definitions

IV.A.3.a. “Brewery” means a source that produces malt beverage and is comprised of emissions units related to brewhouse operations, fermentation, aging or secondary fermentation, and/or packaging operations.

IV.A.3.b. “Packaging operation” means the canning, bottling, or filling of malt beverages into a container. Packaging operations include keg filling. Packaging operations do not include the railcar loading and unloading of beer concentrate shipped off-site for packing.

IV.A.3.c. “Pilot brewery operation” means an operation where total packaging operations are less than 50,000 barrels per year.

IV.A.3.d. “Process loss” means the difference between the quantity of malt beverage sent to packaging and the quantity of malt beverage packaged into a container. Process loss does not include malt beverage in filled containers if the malt beverage is processed after filling to remove or recover ethanol.

IV.A.4. Emission limitations. By May 1, 2019, no owner or operator of a brewery may exceed an average of 6 percent process loss across all packaging operations in a calendar month and 4 percent process loss on a 12-month rolling average during packaging operations.

IV.A.5. Packaging operation work practices

IV.A.5.a. The owner or operator must develop performance objectives and metrics for each packaging operation to reduce spillage and process loss. Process loss records must be summarized annually and compared to performance objectives established by the owner or operator. Process loss records and summaries must be made available to the Division upon request.

IV.A.5.b. The owner or operator must develop and implement an operator training program for employees engaged in packaging operations to understand the operation of the filling lines and minimize breakdowns, spillage, and process loss. The operator training materials must be made available to the Division upon request. At a minimum, the training program must include:

IV.A.5.b.(i) A brewery training manager, coordinator, or equivalent;

IV.A.5.b.(ii) Written standard operating procedures for packaging operations;

IV.A.5.b.(iii) A requirement that initial training be conducted for employees performing packaging operations and more frequently for the following:

IV.A.5.b.(iii)(A) Employees changing packaging operation responsibilities; and

IV.A.5.b.(iii)(B) Startup of new, upgraded, or replaced packaging operations.

IV.A.5.c. The owner or operator must use and maintain packaging operation equipment to reduce container breakage and process loss. For packaging operations, except at pilot brewery operations, this includes, but is not limited to:

IV.A.5.c.(i) Using and maintaining automated filling equipment according to manufacturer recommended procedures or good engineering practices;

IV.A.5.c.(ii) Installing and operating fill level detectors to monitor the liquid fill levels in containers;

IV.A.5.c.(ii) Installing and operating crown inspectors to monitor the condition of crowns and/or caps applied to bottles, if applicable; and

IV.A.5.c.(iv) Utilizing methods to reduce container damage and spillage. This includes, but is not limited to, installing and operating container handling equipment, including smooth glide rails, lubricated conveyors, and variable speed equipment drives.

IV.A.5.d. The owner or operator of pilot brewery operations must use and maintain packaging operation equipment to reduce container breakage and process loss. This includes, but is not limited to:

IV.A.5.d.(i) Maintaining filling equipment according to manufacturer recommended procedures or good engineering practices;

IV.A.5.d.(ii) Monitoring the liquid fill levels in containers; and

IV.A.5.d.(iii) Utilizing methods to reduce container damage and spillage. This includes, but is not limited to, installing and operating container handling equipment, including smooth glide rails, lubricated conveyors, and variable speed equipment drives.

IV.A.6. Wastewater management and treatment. Owners or operators employing microbial and vegetative destruction of VOCs through the land application of wastewater must ensure that the areas where wastewater is applied are areas covered with vegetation at all times when wastewater is applied, except as required following tilling and seeding for crop rotation and field work per standard agricultural practices.

IV.A.7. Recordkeeping

The following records must be kept for a period of five (5) years and made available to the Division upon request:

IV.A.7.a. Monthly records of the percent process loss for packaging operations;

IV.A.7.b. Records necessary to demonstrate compliance with the packaging operation work practice requirements in Section IV.A.5.; and

IV.A.7.c. If applicable, pursuant to Section IV.A.6., monthly and annual records of the amount of wastewater (gallons) sent to the land application site.

V. Control of Emissions from Foam Manufacturing in the 8-hour Ozone Control Area

V.A. Requirements for Foam Product Manufacturing

V.A.1. Applicability

V.A.1.a. Except as provided in Section V.A.2., the requirements of Section V. apply to owners or operators of foam manufacturing operations that existed at a major source of VOC (greater than or equal to 50 tpy VOC) as of January 27, 2020, located in the 8-hour Ozone Control Area.

V.A.1.b. Except as provided in Section V.A.2., the requirements of Section V. apply to owners or operators of foam manufacturing operations that existed at a major source of VOC (greater than or equal to 25 tpy VOC) as of November 7, 2022, located in the 8-hour Ozone Control Area.

V.A.2. Exemptions

Any foam manufacturing operation that uses only non-VOC blowing agents is exempt from this Section V.A.

V.A.3. Definitions

V.A.3.a. “Blowing agent” means any liquid, gaseous or solid substance that alone or in conjunction with other substances is capable of producing a cellular (foam) structure in a polymeric material.

V.A.3.b. “Expandable polystyrene (EPS) beads” means polystyrene beads, particles, or granules, usually less than one-twelfth inch in diameter, that are formulated with a blowing agent (typically 3.5% to 7% of bead weight). When subjected to prescribed heating in an expansion system, the beads puff up, expanding many times their original volume into low density foam globules (called “prepuff” or “puff”) from which a variety of EPS foam products are molded.

V.A.3.c. “Expanded polystyrene (EPS) foam” means a lightweight, foam material, made of polystyrene, from which a variety of common items are made, such as ice-chests, insulation board, protective packaging, and single-use cups.

V.A.3.d. “Foam” means a solid material in a lightweight cellular form (having internal voids or cavities called cells that contain air or a gas) resulting from the introduction or generation of gas bubbles throughout its mass during processing.

V.A.3.e. “Foam manufacturing operation” means any EPS production line, or portion of a production line, which processes raw EPS bead into final molded EPS product. Production line processes include, but are not limited to: pre-expansion, aging (pre-puff), and molding. The manufacturing process ends after the product exits the EPS mold. “Foam manufacturing operation” also means any production line processing methylene diphenyl diisocyanate (MDI), resins, and various hardeners and thickeners into foam products and which results in VOC emissions into the atmosphere. The manufacturing process ends after the product exits the drying tunnel.

V.A.3.f. “Non-VOC blowing agent” means a blowing agent which does not contain VOCs.

V.A.3.g. “Polystyrene” means any grade, class, or type of thermoplastic polymer, alloy, or blend that is composed of at least 80% polymerized styrene by weight.

V.A.3.h. “Raw material” means all polystyrene, polyethylene and polypropylene, and blowing agent used in the manufacture of foam products.

V.A.4. Emission Limitations

V.A.4.a. By May 1, 2022, for sources subject pursuant to Section V.A.1.a. and by May 1, 2024, for sources subject pursuant to Section V.A.1.b., owners and operators of foam manufacturing operations must either

V.A.4.a.(i) Limit VOC emissions from foam manufacturing to 3.0 lbs. per 100 lbs. of total material process, averaged monthly, or

V.A.4.a.(ii) Control VOC emissions from foam manufacturing by 90%. The control device must have a control efficiency of at least 95%.

V.A.5. Work Practices

The owner or operator of any foam manufacturing operation must implement the following work practice requirements at all times to reduce VOC emissions from fugitive sources.

V.A.5.a. Store raw materials in closed, leak-free, labeled containers when not in use.

- V.A.5.b. Cover open containers in a manner that minimizes evaporation into the atmosphere.

V.A.6. Monitoring

- V.A.6.a. The owner or operator of foam manufacturing operations must operate and maintain the control device consistent with the manufacturer's specifications.
- V.A.6.b. By November 1, 2022, and every three (3) years afterward, owners or operator of foam manufacturing operations must conduct a performance test during representative operations using EPA Method 24 (October 7, 2020) to determine VOC content and EPA Method 18, 25, or 25A (November 14, 2018) to determine control efficiency of the emission control equipment.

V.A.7. Recordkeeping

The following records must be kept for a period of five (5) years and made available to the Division upon request

- V.A.7.a. Any records necessary to demonstrate that an exemption in Section V.A.2. applies.
- V.A.7.b. The amount of raw material processed on a daily basis.
- V.A.7.c. The type of blowing agent used.
- V.A.7.d. The amount of blowing agent used on a monthly basis.
- V.A.7.e. The total monthly VOC emissions.
- V.A.7.f. For operators complying with the emission limitation in Section V.A.4.a.(i), the total monthly VOC emissions calculated on a pounds per 100 lbs. of material processed basis.
- V.A.7.g. For operators that use a control device to comply with the emission limitations in Section V.A.4.a.
 - V.A.7.g.(i) A manufacturer guarantee of the control equipment's emission control efficiency to demonstrate compliance with Section V.A.4.
 - V.A.7.g.(ii) The amount of supplementary natural gas combusted in the control device on a monthly basis.
 - V.A.7.g.(iii) Records of performance tests conducted pursuant to Section V.A.6.

V.A.7.h. Records of calendar year VOC emission estimates demonstrating whether the foam manufacturing operation meets or exceeds the applicability threshold in Section V.A.1.

V.A.8. Reporting

V.A.8.a. Performance test protocols required for performance tests under Section V.A.6.b. must be submitted to the Division for review at least thirty (30) days prior to testing and in accordance with AQCC Common Provisions Regulation Section II.C.

V.A.8.b. Beginning in 2025, the owner or operator must submit with the semi-annual report required by the facility's operating permit documentation of compliance with the VOC emission limits in Section V.A.4.a., if applicable.

VI. Control of Emissions from Bakeries in the 8-hour Ozone Control Area

VI.A. Requirements for Bakeries

VI.A.1. Applicability

Beginning May 1, 2023, the requirements of Section VI. apply to owners or operators of bakery operations and bakery recycling that existed at a major source of VOC (greater than or equal to 25 tpy VOC) as of November 7, 2022, located in the 8-hour Ozone Control Area.

VI.A.2. Definitions

VI.A.2.a. "Bakery operation" means the facility and equipment producing flour-based food baked in an oven.

VI.A.2.b. "Bakery oven" means an enclosed compartment supplied with heat used to bake yeast-leavened products including, but not limited to, bread, buns, and rolls.

VI.A.2.c. "Bakery recycling" means the processing of bakery and snack food scrap into animal feed suitable for cattle, swine, and poultry.

VI.A.2.d. "Oven" means a chamber used to bake by means of heat. This does not include proof boxes.

VI.A.2.e. "Proof box" means a warm, humid chamber where yeast-leavened dough is allowed to rise to the volume desired for baking.

VI.A.3. Work practices

VI.A.3.a. Operate and maintain each oven in accordance with the manufacturer's design and operating specifications.

VI.A.3.b. Clean each oven and proof box in accordance with the manufacturer's recommendations and the facility's sanitation standard operating procedures.

VI.A.3.c. Optimize the addition of yeast, fermentation and baking times, and process temperatures to minimize emissions.

VI.A.3.d. Clean bakery recycling dryer and associated ducting in accordance with manufacturer recommendations and the facility's standard operating procedures.

VI.A.3.e. Maintain and operate the oxidizer whenever bakery recycling dryer or bakery oven is in operation, specifically, but not limited to, maintaining the oxidizer temperature in the manufacturer recommended range.

VI.A.3.f. Maintain the oxidizer in accordance with manufacturer recommended maintenance procedures to ensure proper operation of the oxidizer.

VI.A.4. Control requirements

VI.A.4.a. Except as provided in Section VI.A.4.b., the owner or operator of a bakery operation must vent bakery oven emissions to an emission control device with a VOC destruction efficiency of at least 95%.

VI.A.4.b. Bakery operations that are not controlling bakery oven emissions as in Section VI.A.4.a. as of May 1, 2023, must install and operate the required emission control device by May 1, 2025.

VI.A.4.c. The owner or operator of a bakery recycling operation must vent bakery and snack food scrap drying emissions to an emission control device with a VOC destruction efficiency of at least 95%

VI.A.4.d. In the event of a bakery oven control device shutdown during production, the owner or operator must immediately stop sponge production. Once the product in the oven has completed the baking cycle, the owner or operator must shutdown the oven until the control device is operating again.

VI.A.5. Recordkeeping

The following records must be kept for a period of five (5) years and made available to the Division upon request.

VI.A.5.a. Records of the calendar year VOC emission estimates demonstrating whether the bakery operation meets or exceeds the applicability threshold in Section VI.A.1.

VI.A.5.b. Monthly records of flour based food production.

VI.A.5.c. Records of the oven and/or proof box manufacturer design and operating specifications.

VI.A.5.d. Records of the oven and/or proof box manufacturer cleaning recommendations.

VI.A.5.e. Records of the facility's sanitation standard operating procedures.

VI.A.5.f. Records demonstrating compliance with the operating, maintenance, cleaning, and optimization requirements in Section VI.A.3., including records of the manufacturer recommended maintenance procedures.

VI.A.5.g. Records of the oxidizer operating temperature.

VI.A.5.h. Records of the manufacturer guarantee of the control equipment's emission destruction efficiency or a performance test conducted during representative operations in accordance with EPA Method 18, 25, 25A, 2, or 2C (November 14, 2018), whichever is applicable.

VI.A.5.i. Records of production without an operating control device as described in Section VI.A.4.d., including the date and time, the ovens and products running, a description of the problems observed, a description of actions taken to minimize emissions during the event, a description and date of any corrective actions taken, and the name of the individual(s) performing corrective actions.

VII. Repealed

VIII. Control of Emissions from Industrial Waste Facilities in the 8-hour Ozone Control Area

VIII.A. Requirements for solid waste facilities that dispose of oil and gas waste

VIII.A.1. Applicability

Beginning May 1, 2023, the requirements of Section VIII. apply to owners or operators of solid waste facilities that dispose of oil and gas waste that existed at a major source of VOC (greater than or equal to 25 tpy VOC) as of November 7, 2022, located in the 8-hour Ozone Control Area.

Beginning May 1, 2026, the requirements of Section VIII. apply to owners or operators of solid waste facilities that dispose of oil and gas waste that existed at a major source of VOC (greater than or equal to 50 tpy VOC) as of July 24, 2024, located in northern Weld County.

VIII.A.2. Definitions

VIII.A.2.a. “Solid waste” means any garbage, refuse, sludge from a waste treatment plant, water supply treatment plant, air pollution control facility, or other discarded material; including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, or community activities. Solid waste does not include solid or dissolved materials in domestic sewage, or solid or dissolved material in irrigation return flows or industrial discharges which are point sources subject to permits under the provisions of the Colorado Water Control Act, Title 25, Article 8, CRS, or materials handled at facilities licensed pursuant to the provisions of the Radiation Control Act in Title 25, Article 11, CRS. Solid waste does not include materials handled at facilities licensed pursuant to the provisions of radiation control in Title 25, Article 11, CRS; excluded scrap metal that is being recycled; or shredded circuit boards that are being recycled.

VIII.A.2.b. “Solid waste facility” means the location and/or facility at which the deposit and final treatment of solid wastes occur.

VIII.A.2.c. “Oil and gas waste” means drilling fluids, produced waters, and other wastes associated with the exploration, development, or production of oil or natural gas.

VIII.A.3. Work practices

VIII.A.3.a. Combine liquid oil and gas waste with a bulking/solidification material that impeded vaporization.

VIII.A.3.b. Direct bury oil and gas waste solids as soon as possible to limit air emissions.

VIII.A.3.c. Implement good air pollution practices that limit air emissions from oil and gas waste during intra-facility transport of materials to burial locations.

VIII.A.4. Recordkeeping

The following records must be kept for a period of five (5) years and made available to the Division upon request.

VIII.A.4.a. Records of the amount of oil and gas waste processed on a monthly basis by weight or volume.

VIII.A.4.b. Records demonstrating compliance with the work practice requirements in Section VIII.A.3.

VIII.A.5. Reporting. Beginning in 2026, the owner or operator must submit with the semi-annual report required by the facility's operating permit documentation of compliance with the work practices in Section VIII.A.3.

IX. Control of Emissions from Cold Rolling Mills

IX.A. Requirements for cold rolling mills

IX.A.1. Applicability

Beginning May 1, 2025, the requirements of Section IX. apply to owners or operators of cold rolling mills at aluminum sheet manufacturing facilities located at a major source of VOC (greater than or equal to 50 tpy VOC) as of January 27, 2020, in the 8-hour Ozone Control Area.

IX.A.2. Definitions

IX.A.2.a. "Cold rolling mill" means a rolling mill with a preset gap between work rolls used to reduce aluminum sheet thickness.

IX.A.2.b. "Roll coolant" means a lubricant used to cool the work rolls and provide lubrication for the product in contact with the work rolls.

IX.A.3. Standards and work practices

IX.A.3.a. Roll coolant used on the cold rolling mill must consist of lubricant with vapor pressure of less than 0.04 mmHg at 68 degrees Fahrenheit and a boiling point temperature of at least 300 degrees Fahrenheit at 14.7 psia.

IX.A.3.b. Control roll coolant application rates per unit of production using an automated system for ensuring process conditions are maintained at optimum levels.

- IX.A.3.c. Maintain the supplied roll coolant temperature within required temperature ranges (subject to limitations of cooling water temperature) to minimize volatilization.
- IX.A.3.d. Implement spill prevention and other waste reduction measures to ensure that the roll coolant supplied to the system remains within the bounds of the storage, circulation, filtration, and treatment systems.
- IX.A.4. Recordkeeping. The following records must be kept for a period of five (5) years and made available to the Division upon request.
 - IX.A.4.a. Records of calendar year VOC emission estimates demonstrating whether the aluminum sheet manufacturing facility meets or exceeds the applicability threshold in Section IX.A.1.
 - IX.A.4.b. Records such as, but not limited to, safety data sheets for the roll coolant used as stated in Section IX.A.3.
 - IX.A.5. Reporting. Beginning in 2025, the owner or operator must submit with the semi-annual report required by the facility's operating permit documentation of compliance with the roll coolant work practices in Section IX.A.3.

PART C Statements of Basis, Specific Statutory Authority and Purpose

I. April 20, 2023

This Statement of Basis, Specific Statutory Authority, and Purpose complies with the requirements of the State Administrative Procedure Act, § 24-4-101, C.R.S., et seq., the Colorado Air Pollution Prevention and Control Act, § 25-7-101, C.R.S., et seq., and the Air Quality Control Commission's (Commission) Procedural Rules, 5 C.C.R. §1001-1.

Basis

To improve the readability and usability of Regulation Number 7 and Regulation Number 22, the Commission adopted revisions restructuring and reorganizing the parts and sections.

Specific Statutory Authority

The Colorado Air Pollution Prevention and Control Act, § 25-7-101, C.R.S., et seq. (the State Air Act or the Act), specifically § 25-7-103.3, directs rule-making agencies, such as the Commission, to review their rules and consider whether the rule is necessary; whether the rule overlaps or duplicates other rules of the agency or with other federal, state, or local government rules; whether the rule is written in plain language and is easy to understand; whether the rule has achieved the desired intent and whether more or less regulation is necessary; whether the rule can be amended to give more flexibility, reduce regulatory burdens, or reduce unnecessary paperwork or steps while maintaining its benefits; whether the rule is implemented in an efficient and effective manner, including the requirements for the issuance of permits and licenses; whether a cost-benefit analysis was performed by the applicable rule-making agency; and whether the rule is adequate for the protection of the safety, health, and welfare of the state or its residents. Based on this review, the rule-making agency will determine whether the existing rules should be continued in their current form, amended, or repealed.

Purpose

The following section sets forth the Commission's purpose in adopting the revisions to Regulation Number 26. The Commission reorganized Regulation Number 7 into four regulations: Part B became Regulation Number 24; Part C became Regulation Number 25; Part D remained in Regulation Number 7; and Part E became Regulation Number 26. The upstream oil and gas intensity and midstream combustion program provisions currently in Regulation Number 22 moved to Regulation Number 7. The manufacturing sector greenhouse gas provisions in Regulation Number 22 became a new Regulation Number 27. To assist in tracking the history of the regulatory revisions, associated statements of basis and purpose, and restructured location, the Commission provides the following tracking table.

Year of rule adoption	Date of rule adoption	Summary of rule(s) adopted	Regulation Number 7 Section (pre-2019 numbering)	Regulation Number 7 Section (numbering as of 12.2022)	Rule & Section (as of 4.2023)
1995	Dec. 21	Clarify substances that are negligibly reactive VOCs.	Section II.B.	Part A, Section II.B.	Regulations 7 and 24-26, Part A

Year of rule adoption	Date of rule adoption	Summary of rule(s) adopted	Regulation Number 7 Section (pre-2019 numbering)	Regulation Number 7 Section (numbering as of 12.2022)	Rule & Section (as of 4.2023)
1996	Mar. 21	Revisions related to the maintenance demonstration.	Sections I.A.1. through I.A.4.; II.D.; II.E.	Part A, Sections I.A.1. through I.A.4.; II.D.; II.E.	Regulations 7 and 24-26, Part A
1996	Nov. 21	Updated NRVOC list. Removed control of VOC emissions from dry cleaning facilities using perchloroethylene.	Section XII.	NA	NA
1998	Oct. 15	Revisions specific to Gates Rubber Company.	Section II.F.	NA	Regulation 24-25, Part A
2001	Jan. 11	Correct discrepancies in posted versus adopted provisions.	Sections III.C.; IX.L.2.c.(1); X.D.2. through XI.A.3.	Part B, Section I.; Part C, Section I.; Part C, Sections II. through III.	Regulation 24, Part B; Regulation 25, Part B (fkna Part C)
2003	Nov. 20	Repealed provisions establishing a procedure for granting exemptions for de minimis sources and for approving alternative compliance plans.	Sections I.A.2. through I.A.4.; II.D.; II.E.	Part A, Sections I.A.1. through I.A.4.; II.D.; II.E.	Regulations 7 and 24-26, Part A

Year of rule adoption	Date of rule adoption	Summary of rule(s) adopted	Regulation Number 7 Section (pre-2019 numbering)	Regulation Number 7 Section (numbering as of 12.2022)	Rule & Section (as of 4.2023)
2004	Mar. 12	Revisions adopted in conjunction with the early action compact ozone action plan – control of emissions from condensate operation at oil and gas facilities, emissions from internal combustion engines, emissions from gas processing plants, and emissions from oil and gas operations dehydrators.	Sections I.A.; I.B.; XII.; XVI.	Part A, Section I.A.; Part A, Section I.B.; Part D, Section I.; Part E, Section I.	Regulations 7 and 24-26, Part A
2004	Dec. 16	Revisions adopted in response to EPA comments (re practical enforceability) on the ozone action plan adopted 3/2004.	Sections I.A.; II.A.; XII.; XVI.;	Part A, Section I.A.; Part A, Section II.A.; Part D, Section I.; Part E, Section I.	Regulations 7 and 24-26, Part A
2006	Dec. 17	Expanding oil and gas condensate tank emission controls.	Section XII.	Part D, Section I.	Regulation Number 7, Part B (fkna Part D)

Year of rule adoption	Date of rule adoption	Summary of rule(s) adopted	Regulation Number 7 Section (pre-2019 numbering)	Regulation Number 7 Section (numbering as of 12.2022)	Rule & Section (as of 4.2023)
2006	Dec. 17	Reduce emissions from oil and gas operations and natural gas fired engines.	Sections I.A.1.b.; XVII.	Part A, Section I.A.; Part D., Section II. & Part E. Section I. (for engines)	Regulation Number 7, Part A and Part B (fkna Part D); Regulation 26, Part A and Part B (fkna Part E)Part E)

Year of rule adoption	Date of rule adoption	Summary of rule(s) adopted	Regulation Number 7 Section (pre-2019 numbering)	Regulation Number 7 Section (numbering as of 12.2022)	Rule & Section (as of 4.2023)
2008	Dec. 12	Expand VOC RACT requirements for 100 tpy sources and clarify how RACT requirements in Regulation Numbers 3 and 7 interact in the ozone nonattainment area. Make typographical and formatting changes. Revise oil and gas condensate tank and pneumatic controller requirements.	Title; Sections I.; II.; VI. through XIII.; XVII.; XVIII.; and Appendices A through F	Part A, Section I.; Part A, Section II.; Part B, Sections IV. through VI. & Part C, Sections I. through IV. & Part D, Section I.; Part D, Section II. and Part E, Section I. (for engines); Part D, Section III.; Part A, Appendix A. & Part B, Appendices B and C & Part C, Appendices D and E (formerly Appendix F)	Regulation 24, Part B; Regulation 25, Part B (fkna Part C); Regulation Number 7, Part B (fkna Part D); Regulation 26, Part B (fkna Part E)

Year of rule adoption	Date of rule adoption	Summary of rule(s) adopted	Regulation Number 7 Section (pre-2019 numbering)	Regulation Number 7 Section (numbering as of 12.2022)	Rule & Section (as of 4.2023)
2011	Jan. 7	Include engine requirements in the Regional Haze SIP.	Outline; Sections I.; XVII.	Part A, Section I.; Part E, Section I.	Regulation 26, Part B (fkna Part E)
2012	Dec. 20	Address EPA comments on the June 2009 submittal. Revise state-only requirements for consistency.	Sections II.; XII.; XVII.	Part A, Section II.; Part D, Section I.; Part D, Section II.	Regulation Number 7, Part B (fkna Part D)
2014	Feb. 23	Adopt additional oil and gas emission reduction requirements – auto-igniters, expand condensate tank controls, limit storage tank venting, expand dehydrator control, establish leak detection and repair program, limit venting during well maintenance and liquids unloading, expand pneumatic controller requirements.	Sections II.; XVII.; XVIII.	Part A, Section II.; Part D, Section II.; Part D, Section III.	Regulation Number 7, Part B (fkna Part D)

Year of rule adoption	Date of rule adoption	Summary of rule(s) adopted	Regulation Number 7 Section (pre-2019 numbering)	Regulation Number 7 Section (numbering as of 12.2022)	Rule & Section (as of 4.2023)
2016	Nov. 17	Adopt RACT requirements for industrial cleaning solvents, lithographic and letterpress printing, and specific major sources. Including existing combustion device auto-igniter and storage tank inspection requirements in the SIP. Adopting major source combustion equipment combustion process adjustment requirements and incorporate by reference NSPS and NESHAP for specific major sources.	Sections I.; X.; XII.; XIII.; XVI.; XIX.	Part A, Section I.; Part C, Section II.; Part D, Section I.; Part C, Section IV.; Part C, Section V.; Part E, Section III.	Regulation 25, Part B (fkna Part C); Regulation Number 7, Part B (fkna Part D); Regulation 26, Part B (fkna Part E)
2017	Nov. 16	Adopt provisions based on recommendations in EPA's Oil and Gas Control Techniques Guideline. Revise state-only requirements for consistency.	Sections II.; XII.; XVII.; XVIII.	Part A, Section II.; Part D, Section I.; Part D, Section II.; Part D, Section III.	Regulation Number 7, Part B (fkna Part D)
2018	July 19	Adopt requirements for existing major source boilers, turbines, lightweight aggregate kilns, glass melting furnaces, engines.	Sections XVI.; XIX.	Part E, Section II., Part E, Section III.	Regulation 26, Part B (fkna Part E)

Year of rule adoption	Date of rule adoption	Summary of rule(s) adopted	Regulation Number 7 Section (pre-2019 numbering)	Regulation Number 7 Section (numbering as of 12.2022)	Rule & Section (as of 4.2023)
2018	Nov. 15	Adopt requirements for major source breweries and wood furniture manufacturing. Address EPA concerns with requirements for industrial cleaning solvents, metal furniture surface coating, and miscellaneous metal surface coating. Updated incorporation by reference dates.	Sections I.; II.; VI.; VIII.; IX.; X.; XII.; XIII.; XVI.; XVII.; XIX.; XX.; XXI.	Part A, Section I.; Part A, Section II.; Part B, Section IV.; Part B, Section VI.; Part C, Section I.; Part C, Section X.; Part D, Section I.; Part C, Section IV.; Part C, Section V.; Part D, Section II.; Part E, Section III.; Part E, Section IV. Part F	Regulation 24, Part B; Regulation 25, Part B (fkna Part C); Regulation Number 7, Part B (fkna Part D); Regulation 26, Part B (fkna Part E); Regulation 23 (fkna Part F)

Year of rule adoption	Date of rule adoption	Summary of rule(s) adopted	Regulation Number 7 Section (pre-2019 numbering)	Regulation Number 7 Section (numbering as of 12.2022)	Rule & Section (as of 4.2023)
2019	Dec. 19	Reorganized into Parts A through F. Replaced the SIP system-wide condensate tank control program with a fixed threshold storage tank control program. Increased state-only, state-wide storage tank controls. Adopted oil and gas storage tank measurement system, hydrocarbon liquids loadout, leak detection and repair, well plugging, and pneumatic controller requirements. Adopted an oil and gas transmission and storage segment methane intensity program. Adopted an annual oil and gas inventory program. Expanded SIP requirements to 50 tpy sources. Aligned gasoline tank truck testing requirements with federal requirements as SIP clean-up.	Sections I. through XX. and Appendices A through F	(see reorganization cross walk)	

Year of rule adoption	Date of rule adoption	Summary of rule(s) adopted	Regulation Number 7 Section (pre-2019 numbering)	Regulation Number 7 Section (numbering as of 12.2022)	Rule & Section (as of 4.2023)
2020	Sept. 23	Adopted requirements for natural gas fired 1,000 horsepower engines. Adopted flowback vessel control requirements and pre- and early-production monitoring requirements. Expanded hydrocarbon liquids loadout requirements to class II disposal well facilities.		Part D, Sections II.; IV.; V.; VI.; Part E, Section I.	Regulation Number 7, Part B (fkna Part D); Regulation 26, Part B (fkna Part E)
2020	Dec. 18	Adopted requirements for major source foam manufacturing, boilers, turbines, landfill and biogas fired engines, and wood surface coating.		Part D, Section II.; Part E, Sections II.; IV.; V.	Regulation Number 7, Part B (fkna Part D); Regulation 26, Part B (fkna Part E)
2021	Feb. 18	Adopted non-emitting pneumatic controller requirements for new facilities and existing pneumatic controller retrofit requirements for existing facilities.		Part D, Section III.	Regulation Number 7, Part B (fkna Part D)

Year of rule adoption	Date of rule adoption	Summary of rule(s) adopted	Regulation Number 7 Section (pre-2019 numbering)	Regulation Number 7 Section (numbering as of 12.2022)	Rule & Section (as of 4.2023)
2021	July 16	Adopted requirements for metal parts surface coating and major source process heaters.		Part C, Section I.; Part D, Section III.; Part E, Section II.	Regulation 25, Part B (fkna Part C); Regulation Number 7, Part B (fkna Part D); Regulation 26, Part B (fkna Part E)
2021	Dec. 17	Adopted SIP revisions to address EPA concerns with the EPA Oil and Gas CTG. Adopted oil and gas combustion device performance testing requirements. Expanded reciprocating compressor rod packing, leak detection and repair, and pneumatic controller requirements at natural gas processing plants. Expanded leak detection and repair, separator, and well maintenance requirements. Adopted pigging and blowdown requirements.		Part D, Sections I., II., III., V., VI.	Regulation Number 7, Part B (fkna Part D)

Year of rule adoption	Date of rule adoption	Summary of rule(s) adopted	Regulation Number 7 Section (pre-2019 numbering)	Regulation Number 7 Section (numbering as of 12.2022)	Rule & Section (as of 4.2023)
2022	Dec. 15	Adopted requirements for major source combustion equipment, wood coating, solvent use, bakery operation, digital printing, poultry waste processing, oil stabilization facilities, class II injection well facilities, and industrial waste; included state only provisions as SIP strengthening measures; clarified the applicability of requirements to newly classified ozone nonattainment areas; included requirements for motor vehicle materials and automotive coatings; expanded gasoline tank truck testing requirements.		Part E, Sections I., II., III., VI., VII., and VIII., Part C., Sections I., II., and IV., and Part D, Section II.; Part D, Sections II., ; Part A, Sections I. and II.; Part C, Section I.; and Part B, Section IV.	Regulation 24, Part B (fkna Part B); Regulation 25, Part B (fkna Part C); Regulation Number 7, Part B (fkna Part D); Regulation 26, Part B (fkna Part E)

The Commission also made typographical, grammatical, and formatting corrections throughout the regulations.

Incorporation by Reference

The Commission will update regulatory references as needed as opportunities arrive.

Additional Considerations

These revisions are administrative in nature and, therefore, do not exceed or differ from the requirement of the federal act or rules. Therefore, § 25-7-110.5(5)(a) does not apply.

Findings of Fact

To the extent that § 25-7-110.8, C.R.S., requirements apply to this rulemaking, and after considering all the information in the record, the Commission hereby makes the determination that:

- (I) These rules are based upon reasonably available, validated, reviewed, and sound scientific methodologies, and the Commission has considered all information submitted by interested parties.
- (II) Evidence in the record supports the finding that the rules shall result in a demonstrable reduction of greenhouse gas and VOC emissions.
- (III) Evidence in the record supports the finding that the rules shall bring about reductions in risks to human health and the environment that justify the costs to implement and comply with the rules.
- (IV) The rules are the most cost-effective alternative to achieve the necessary reduction in air pollution and provide the regulated entity flexibility.
- (V) The selected regulatory alternative will maximize the air quality benefits of regulation in the most cost-effective manner.

II. December 15, 2023 (Revisions to Part A, Section I.C.; and Part B, Sections I.D.5., I.D.6., II., III.D., and V.)

This Statement of Basis, Specific Statutory Authority, and Purpose complies with the requirements of the State Administrative Procedure Act, § 24-4-101, C.R.S., et seq., the Colorado Air Pollution Prevention and Control Act, § 25-7-101, C.R.S., et seq., and the Air Quality Control Commission's (Commission) Procedural Rules, 5 C.C.R. §1001-1.

Basis

On October 7, 2022, EPA reclassified the Denver Metro/North Front Range (DM/NFR) to severe for the 2008 8-hour Ozone National Ambient Air Quality Standard of 75 parts per billion (ppb) (2008 ozone NAAQS), after 2019-2021 ozone data failed to show attainment. See 86 Fed. Reg. 60926. Separately, EPA has also designated the DM/NFR as marginal nonattainment for the 2015 ozone NAAQS of 70 ppb, effective August 3, 2018 (83 Fed. Reg. 25776 (June 4, 2018)). On November 30, 2021, EPA expanded the boundary of the 2015 ozone nonattainment area to include all of Weld County, effective December 30, 2021 (86 Fed. Reg. 67864). On October 7, 2022, EPA reclassified the DM/NFR and northern Weld County to moderate, after 2019-2021 ozone data failed to show attainment. See 86 Fed. Reg. 60897.

To ensure progress towards attainment of the 2008 and 2015 ozone NAAQS and respond to Colorado's requirements under the Clean Air Act, the Commission adopted revisions to include reasonably available control technology (RACT) for major sources of volatile organic compounds (VOC) and nitrogen oxides (NOx) in the nonattainment areas, specifically expanding requirements for foam manufacturing, modifications to requirements for landfill and biogas fuel fired engines, modifications to requirements for process heaters, and a requirement for additional major source RACT analyses. The Commission also adopted additional, state-only, state-wide requirements for stationary combustion engines.

Specific Statutory Authority

The State Air Act, specifically § 25-7-105(1), directs the Commission to promulgate such rules and regulations as are consistent with the legislative declaration set forth in § 25-7-102 and that are necessary for the proper implementation and administration of Article 7. The Act broadly defines air pollutant to include essentially any gas emitted into the atmosphere (and, as such, includes VOC, NOx, methane and other hydrocarbons) and provides the Commission broad authority to regulate air pollutants. Section 105(1)(a)(I) directs the Commission to adopt a state implementation plan (SIP) to attain the NAAQS. § 25-7-106 provides the Commission maximum flexibility in developing an effective air quality program and promulgating such combination of regulations as may be necessary or desirable to carry out that program. § 25-7-106 also authorizes the Commission to promulgate emission control regulations applicable to the entire state, specified areas or zones, or a specified class of pollution. § 25-7-106(6) further authorizes the Commission to require owners and operators of any air pollution source to monitor, record, and report information. §§ 25-7-109(1)(a) and (2) of the Act authorize the Commission to promulgate regulations requiring effective and practical air pollution controls for significant sources and categories of sources and emission control regulations pertaining to nitrogen oxides and hydrocarbons.

Purpose

The following section sets forth the Commission's purpose in adopting the revisions to Regulation Number 26, and includes the technological and scientific rationale for the adoption of the revisions.

Stationary Combustion Engines

In 2020, the Commission adopted state-wide, state-only requirements to minimize emissions from natural gas-fired reciprocating internal combustion engines greater than or equal to 1,000 horsepower (hp). At that time, the Commission requested that the Division consider evaluating strategies to increase the electrification of engines, lower emissions standards for engines, and possible controls applicable to smaller engines. In response to this directive and to further address NOx emissions affecting Colorado's ability to meet the ozone NAAQS, as well as other state goals, the Commission adopted additional NOx emission limits for natural gas-fired engines greater than or equal to 100 hp and diesel engines greater than or equal to 500 hp.

For engines placed in service, modified, or relocated after January 30, 2024, the Commission adopted, generally, more stringent NOx standards. The Commission also intends that any engines subject to a more stringent standard under a permit must still comply with that more stringent limit. The Commission adopted varying timing requirements for owners or operators to meet the emission standards, based on the location of subject engines inside and outside of the 8-hour ozone control area. Owners or operators with any natural gas-fired engines in the 8-hour ozone control area are subject to a more aggressive timeline, which requires 100% of engines inside the 8-hour ozone control area to meet the emission standards by May 1, 2027, and 100% of engines outside the 8-hour ozone control area meet the emission standards by May 1, 2029. Operators with no engines inside the 8-hour ozone control area must follow the second timeline and meet the standards of at least 20% of engines each year from 2024 to 2029. Owners and operators of subject diesel and dual-fuel engines statewide are subject to a different phase in schedule of 50% by May 1, 2027 and 100% by May 1, 2029. This phase-in schedule was adopted to allow operators with very small fleets, potentially a single engine, adequate time to plan for retrofits or replacements of these engines. The Commission adopted requirements for owners and operators to submit permit applications to incorporate the applicable emissions standards under Section I.D.6.b. For engines that must submit a general construction permit modification application or a Title V minor modification application pursuant to Section I.D.6.b.(vi)(B), the Commission intends that the emission limits in the applicable permit existing as of the date of submittal of the permit modification will continue to be the enforceable emission limits until the applicable compliance deadline established in Section I.D.6.b.(vii) unless the permit modification application requests the emission limits in Section I.D.6.b. be the enforceable limits earlier than the compliance deadline established in Section I.D.6.b.(vii). In the event that the Division encounters delays in the Division's permitting process for timely, complete permit applications, the Commission intends that the Division will extend the compliance timelines for retrofits, replacements, and performance testing as necessary. The intent of Section I.D.6.a.(iv) "one-time only replacement" applies to a specific engine at a specific site i.e. each specific engine that was in service as of January 30, 2024 at a site may only be replaced one time.

The Commission understands a small subset (~15 engines statewide) of lean burn engines placed in service, modified, or relocated on or before January 30, 2024, may not be able to feasibly obtain the NOx emission standard in Section I.D.6.b.(ii) (referencing the emissions standards in Table 5). To address this issue, the Commission has provided a mechanism through which owners or operators of those engines can apply to the Division for an alternative NOx emission standard if the owner or operator can demonstrate that it is not technically or economically feasible to comply with the proposed NOx emission standard in Section I.D.6.b.(ii). The Division must grant an

alternative emission standard where the Division determines that a satisfactory source-specific demonstration has been made. In adopting this provision, it is not the Commission's intent to require RACT in attainment areas (other than as required by Regulation Number 3, Part B, Section III.D.2.); rather, the Commission intends that the Division apply a RACT-like analysis in determining an appropriate alternative emission standard.

In evaluating economic infeasibility as part of the source-specific demonstration, the Division must consider (1) the cost-effectiveness (in cost per ton of pollutant reduced) of undertaking efforts to achieve the standard using generally accepted methods for determining cost per ton reduced and (2) recent and relevant cost-effectiveness thresholds used by the Commission and/or Division with consideration that the source is located in an ozone attainment area. The Division will set an appropriate alternative emission standard by applying the RACT standard (i.e., control technology that will achieve the maximum degree of emission control that a particular source is capable of meeting and that is reasonably available considering technological and economic feasibility). The Commission does not intend to make alternative NO_x emission standards available to engines located in northern Weld County or the 8-Hour ozone control area.

The Commission intends that the emission standards in Table 5 are a gram per horsepower-hour (g/hp-hr) limit based on appropriate averaging times. The Commission also intends that operators demonstrate compliance with the certification and recordkeeping requirements through the performance testing results required by Section I.D.6.c. and the portable analyzer results obtained in accordance with Section I.D.6.d., using the appropriate averaging times. Owners and operators that are subject to more frequent performance testing as part of an existing permit condition must continue to comply with any applicable permit requirement.

The Commission understands that the affected engines in the midstream sector will be subject to a future scheduled rulemaking based on recommendations of the Midstream Steering Committee (MSC) per a regulatory process the Commission adopted in December 2021. The result of the MSC will be a rulemaking in 2024 where operators will commit to reducing GHG emissions from the sector with NO_x reductions being a “co-benefit” as operators will either electrify, shut down, and/or otherwise reduce emissions from some portion of their natural gas-fired engines to meet 2030 GHG emission targets. The Commission does not intend that operators take on the cost of retrofits and replacements under this rule just to later either replace an engine with an electric unit or to wholly remove an engine from service. However, because the Commission cannot forecast the outcome of the 2024 rulemaking, it directs the Division to evaluate, in consultation with midstream operators and other stakeholders, if Regulation Number 26 should be opened concurrently with the 2024 MSC rulemaking timeline to address any conflicts between this rule and the 2024 MSC proposed rules that will be developed by the Division.

The Commission understands that the Division currently collects data on engines that operate at <100 HP as part of the Oil and Natural Gas Annual Emission Inventory Reporting (ONGAIER) data system on an annual basis. The Commission directs the Division to evaluate the existing ONGAIER dataset related to these engines and determine whether there are technically and economically feasible emission control strategies that should be considered by the Commission in a future rulemaking.

Major Source RACT

Due to the reclassifications to severe and moderate, Colorado must submit revisions to its SIP to address the Clean Air Act’s (CAA) ozone nonattainment area requirements, as set forth in CAA §§ 172, 182(b), 182(d), and the final SIP Requirements Rules. Severe SIPs must include provisions that require the implementation of RACT for major sources of VOC and/or NO_x (i.e., sources that emit or have the potential to emit 25 tpy or more) and for each category of VOC sources covered by a Control Technique Guideline (CTG) for which Colorado has sources in the nonattainment area.

Therefore, to address the severe nonattainment area requirements under CAA § 182(d), the Commission adopted revisions to Regulation Number 26 to include RACT requirements in Colorado’s ozone SIP for 25 tpy major sources of VOC and/or NO_x including expanding the foam manufacturing operations. In response to EPA concerns and limited disapproval, the Commission also adopted revised requirements for process heaters, landfill and biogas engines, a coil coating facility, and periodic reporting.

Foam Manufacturing

In response to the DM/NFR being reclassified to severe nonattainment under the 2008 ozone NAAQS the Commission adopted revisions to Regulation 26 to expand VOC control requirements initially adopted in December 2020 to foam manufacturing operations with VOC emissions greater than or equal to 25 tons per year (tpy). These provisions include work practice, monitoring, and recordkeeping requirements for foam manufacturing operations.

Landfill and Biogas Fuel Fired Engines

In 2020, the Commission expanded the NO_x emission limit requirements for compression ignition reciprocating internal combustion engines (RICE) and combustion process adjustment requirements for stationary RICE to landfill and biogas fired engines at major sources (50 tpy). Considering the potentially subject sources at that time, the Commission adopted the 2.0 g/hp-hr NO_x emission limit in EPA's NSPS JJJJ for landfill/digester gas fired engines. In reviewing the adopted requirements and considering the remaining subject engines, EPA has raised concerns with the NO_x emission limit and testing requirements. The Commission re-reviewed the subject engines, several having since been removed from service, and is adopting a revised NO_x emission limit of 1.5 g/hp-hr for these engines. The Commission also expanded the periodic performance testing requirements currently applicable to other subject major source engines.

Process Heaters

In 2021, the Commission adopted additional requirements for refinery fuel fired process heaters at major NO_x sources. In response to EPA's comments and concerns and to allow for further evaluation, the Commission removed the NO_x emission limits for these heaters. The process heaters will continue to be subject to combustion process adjustment requirements, as they have been since the 2021 revisions, and performance testing requirements. The Commission directs the Division to return to the Commission with proposed revisions in the future related to these heaters following further useful input from EPA, if necessary and appropriate.

Golden Aluminum (coil coating facility)

In 2020, the Commission evaluated the NO_x emission points at Golden Aluminum as it became a major source of NO_x emissions under the 50 tpy serious major stationary source threshold. Golden Aluminum is a coil coating operation and, therefore, subject to the coil coating provisions in Regulation Number 25 (formerly in Regulation Number 7, Part C), which were adopted many years ago based on the recommendations in EPA's corresponding coil coating CTG. EPA is taking issue with Colorado's reliance on EPA's long-standing interpretation and position that states may utilize EPA's control techniques guidelines (CTG) recommendations and definition of the subject VOC source category when revising the state's SIP to include provisions that require the implementation of RACT for that particular VOC source category (i.e., SIP RACT). Specifically, EPA has proposed a limited disapproval because Colorado did not evaluate VOC emission points at this coil coating operation that were also not addressed in EPA's coil coating CTG. Therefore, the Commission adopted a requirement for Golden Aluminum to conduct and submit a RACT analysis to the Division for further evaluation of the VOC-emitting points at issue in EPA's proposed disapproval.

The Commission also made typographical, grammatical, and formatting corrections throughout the regulation.

Incorporation by Reference

The Commission will update regulatory references as needed as opportunities arrive.

Additional Considerations

Colorado must revise Colorado's ozone SIP to address the severe ozone nonattainment area requirements. The CAA does not expressly address all of the provisions adopted by the Commission. Rather, federal law establishes the ozone NAAQS and requires Colorado to develop a SIP adequate to attain the NAAQS. Therefore, the Commission adopted certain revisions to Regulation Number 26 to satisfy Colorado's nonattainment area obligations and further achieve reductions of ozone precursor emissions. These revisions do not exceed or differ from the federal act due to state flexibility in determining what control strategies to implement to reduce emissions. However, where the proposal may differ from federal rules under the federal act, in accordance with § 25-7-110.5(5)(b), CRS, the Commission determines:

- (I) The revisions to Regulation Number 26 address foam manufacturing, landfill biogas fuel fired engines, a coil coating operation, and process heaters. NSPS T, MACT DDDDD, MACT ZZZZ, and MACT SSSS may also apply to and the above listed equipment and operations. However, the revisions to Regulation Number 26 apply on a broader basis.

- (II) The federal rules discussed in (I) are primarily technology-based in that they largely prescribe the use of specific technologies or work practices to comply.
- (III) The CAA establishes the 2008 and 2015 ozone NAAQS and requires Colorado to develop SIP revisions that will ensure attainment of the NAAQS. The ozone NAAQS was not determined taking into account concerns unique to Colorado. Similarly, EPA develops NSPS or NESHAP considering national information and data, not Colorado specific issues or concerns. In addition, Colorado cannot rely exclusively on a federally enforceable permit or federally enforceable NSPS or NESHAP to satisfy Colorado's ozone nonattainment area RACT obligations. Instead, Colorado can adopt applicable provisions into its SIP directly, as the Commission has done here.
- (IV) In addition to the 2008 NAAQS, Colorado must also comply with the lower 2015 ozone NAAQS. These current revisions may improve the ability of the regulated community to comply with new requirements needed to attain the lower NAAQS insofar as RACT analyses and efforts conducted to support the revisions adopted by the Commission may prevent or reduce the need to conduct additional RACT analyses for the more stringent NAAQS.
- (V) EPA has established Colorado's SIP RACT implementation deadlines. There is no timing issue that might justify changing the time frame for implementation of federal requirements.
- (VI) The revisions to Regulation Number 26 strengthen Colorado's SIP. These sections currently address emissions from foam manufacturing, landfill and biogas fuel fired engines, a coil coating operation, and process heaters, while allowing for continued growth of Colorado's industry.
- (VII) The revisions to Regulation Number 26 establish reasonable equity for owners and operators subject to these rules by providing the same standards for similarly situated and sized sources.
- (VIII) If EPA does not approve Colorado's SIP, EPA may promulgate a Federal Implementation Plan; thus potentially determining RACT for Colorado's sources. This outcome may subject others to increased costs.
- (IX) Where necessary, the revisions to Regulation Number 26 include minimal monitoring, recordkeeping, and reporting requirements that correlate, where possible, to similar federal or state requirements.

- (X) Demonstrated technology is available to comply with the revisions to Regulation Number 26. Some of the revisions expand upon requirements already applicable. The revisions concerning major sources of NO_x generally reflect current emission controls and work practices.
- (XI) As set forth in the Economic Impact Analysis, the revisions to Regulation Number 26 will reduce emissions in a cost-effective manner.
- (XII) Alternative rules could also provide reductions in ozone, VOC, and NO_x to help to attain the NAAQS. However, a no action alternative would very likely result in an unapprovable SIP.

Findings of Fact

To the extent that § 25-7-110.8, C.R.S., requirements apply to this rulemaking, and after considering all the information in the record, the Commission hereby makes the determination that

- (I) These rules are based upon reasonably available, validated, reviewed, and sound scientific methodologies, and the Commission has considered all information submitted by interested parties.
- (II) Evidence in the record supports the finding that the rules shall result in a demonstrable reduction of greenhouse gas and VOC emissions.
- (III) Evidence in the record supports the finding that the rules shall bring about reductions in risks to human health and the environment that justify the costs to implement and comply with the rules.
- (IV) The rules are the most cost-effective alternative to achieve the necessary reduction in air pollution and provide the regulated entity flexibility.
- (V) The selected regulatory alternative will maximize the air quality benefits of regulation in the most cost-effective manner.

III. December 18-20, 2024 (Revisions to Part B, Sections I.D.4.c., I.D.6.a., II.A.4.g., II.A.5.b., II.A.8.c., III.D., V.A.8.b., and IX.)

This Statement of Basis, Specific Statutory Authority, and Purpose complies with the requirements of the State Administrative Procedure Act, § 24-4-101, C.R.S., et seq., the Colorado Air Pollution Prevention and Control Act, § 25-7-101, C.R.S., et seq., and the Air Quality Control Commission's (Commission) Procedural Rules, 5 C.C.R. §1001-1.

Basis

The Commission adopted revisions in response to EPA actions on Colorado's state implementation plan (SIP) submissions. Specifically, a disapproval of the provisions requiring implementation of reasonably available control technology (RACT) for refinery fueled process heaters at major NOx stationary sources and a cold mill coil coating operation and a disapproval related to recordkeeping and reporting requirements. See 88 Fed. Reg. 85511 (Dec. 8, 2023); 88 Fed. Reg. 29827 (May 9, 2023).

Specific Statutory Authority

The State Air Act, specifically § 25-7-105(1), directs the Commission to promulgate such rules and regulations as are consistent with the legislative declaration set forth in § 25-7-102 and that are necessary for the proper implementation and administration of Article 7. The Act broadly defines air pollutant to include essentially any gas emitted into the atmosphere (and, as such, includes VOC, NOx, methane and other hydrocarbons) and provides the Commission broad authority to regulate air pollutants. Section 105(1)(a)(I) directs the Commission to adopt a state implementation plan (SIP) to attain the NAAQS. § 25-7-106 provides the Commission maximum flexibility in developing an effective air quality program and promulgating such combination of regulations as may be necessary or desirable to carry out that program. § 25-7-106 also authorizes the Commission to promulgate emission control regulations applicable to the entire state, specified areas or zones, or a specified class of pollution. § 25-7-106(6) further authorizes the Commission to require owners and operators of any air pollution source to monitor, record, and report information. §§ 25-7-109(1)(a) and (2) of the Act authorize the Commission to promulgate regulations requiring effective and practical air pollution controls for significant sources and categories of sources and emission control regulations pertaining to nitrogen oxides and hydrocarbons.

Purpose

The following section sets forth the Commission's purpose in adopting the revisions to Regulation Number 26, and includes the technological and scientific rationale for the adoption of the revisions.

Midstream segment engines

In December 2023, the Commission adopted NOx emission standards for stationary rich burn natural gas fired reciprocating internal combustion engines with a manufacturer's design rate greater than or equal to 100 horsepower but less than 1000 horsepower and lean burn natural gas fired reciprocating internal combustion engines with a manufacturer's design rate greater than or equal to 250 horsepower but less than 1000 horsepower. Owner or operators of engines placed in service, modified, or relocated before January 30, 2024, had to achieve an increasing percentage of compliance starting by May 1, 2025, and through May 1, 2029. During that rulemaking, the Commission recognized that engines in the midstream segment could be impacted by both the Regulation Number 26 requirements as well as the future midstream rulemaking and stated that the Commission did "not intend that operators take on the cost of retrofits and replacements under this rule just to later either replace an engine with an electric unit or to wholly remove an engine from service." The Commission directed the Division to evaluate whether the Regulation Number 26 requirements should be revised during the midstream rulemaking to address potentially conflicting provisions. Therefore, the Commission now adopts an exemption for engines in the midstream segment from the Regulation Number 26 requirements should the owner or operator remove or electrify the engine pursuant to the midstream segment emission reduction program.

Engine clean-up

In December 2022, the Commission adopted SIP-strengthening measures that incorporated requirements for specific 1,000 hp engines (requirements adopted in 2020). Table B in Part B, Section I.D.4.c. was based on operator engine reporting that began in 2021. As part of that reporting, operators identified engines fleet-wide that could achieve emission reductions through permitted emissions reductions, installation of additional controls, engine replacement, and shutdowns. Table B included engines that were scheduled for shutdown in 2021; however, some of the reports received by the Division were not accurate. The revisions to Table B reflect that some engines were listed in error.

Process Heaters

In 2021, the Commission adopted additional requirements for refinery fuel fired process heaters at major NO_x sources. In response to EPA's comments and concerns and to allow for further evaluation, the Commission removed the NO_x emissions limits for these heaters in 2023. The process heaters continued to be subject to combustion process adjustment requirements, as they have been since the 2021 revisions. Following further evaluation, the Commission now adopts operational and/or NO_x emissions limits for these heaters. Additional changes were made to clarify that the limits in Part B, Tables 2, 3, and 4 only apply to process heaters using the primary fuel type that provides the majority of the heat input to the process heater on a calendar year basis. The operational practices identified in Part B, Table 3 identify existing pollution reduction equipment on the listed heaters as of October 2024 as RACT. The identification of existing operational practices is not intended to limit or prevent the voluntary installation and operation of new or different pollution control equipment that further reduce emissions beyond the RACT requirements established in this rule. Additional voluntary reductions might, depending on the circumstances, qualify for emission reduction credits if all other requirements of Regulation Number 3, Part A, Section V. are met.

Golden Aluminum (coil coating facility)

In 2020, the Commission evaluated the NO_x emission points at Golden Aluminum as it became a major source of NO_x emissions under the 50 tpy serious major stationary source threshold. Golden Aluminum is a coil coating operation and, therefore, subject to the coil coating provisions in Regulation Number 25 (formerly in Regulation Number 7, Part C), which were adopted many years ago based on the recommendations in EPA's corresponding coil coating CTG. EPA raised concerns with Colorado's reliance on EPA's long-standing interpretation and position that states may utilize EPA's control techniques guidelines (CTG) recommendations and definition of the subject VOC source category when revising the state's SIP to include provisions that require the implementation of RACT for that particular VOC source category (i.e., SIP RACT). EPA published a limited disapproval because Colorado did not evaluate certain VOC emission points at this coil coating operation. See 88 Fed. Reg. 85511 (Dec. 8, 2023). Therefore, in 2023, the Commission adopted a requirement for Golden Aluminum to conduct and submit a RACT analysis to the Division for further evaluation of the VOC-emitting points at issue in EPA's disapproval. Following the submission of that RACT analysis, the Commission now adopts requirements for owners or operators of cold rolling mill roll coolant at aluminum sheet manufacturing facilities located at a major source of VOC.

Periodic Reporting

To address EPA's concern with the lack of specified reporting and to avoid federal sanctions the Commission included requirements for major sources to include compliance demonstrations with the semi-annual reporting requirements in the operating permit, specifically for certain combustion equipment at major NOx sources and major VOC source foam manufacturing operations. See 88 Fed. Reg. 29827 (May 9, 2023) (EPA's limited disapproval, discussing EPA's perceived lack of periodic reporting sufficient to determine compliance by regulated entities). Submission of records maintained pursuant to Part B, Section II.A.7.f. will generally constitute sufficient "documentation" of compliance with the combustion process adjustment requirements pursuant to Part B, Section II.A.8.c., but the Division may require the submission of additional documentation if deemed necessary to demonstrate compliance.

The Commission also made typographical, grammatical, and formatting corrections throughout the regulation.

Incorporation by Reference

The Commission will update regulatory references as needed as opportunities arrive.

Additional Considerations

Colorado must revise Colorado's ozone SIP to address the severe ozone nonattainment area requirements. The CAA does not expressly address all of the provisions adopted by the Commission. Rather, federal law establishes the ozone NAAQS and requires Colorado to develop a SIP adequate to attain the NAAQS. Therefore, the Commission adopted certain revisions to Regulation Number 26 to satisfy Colorado's nonattainment area obligations and further achieve reductions of ozone precursor emissions. These revisions do not exceed or differ from the federal act due to state flexibility in determining what control strategies to implement to reduce emissions. However, where the proposal may differ from federal rules under the federal act, in accordance with § 25-7-110.5(5)(b), CRS, the Commission determines:

- (I) The revisions to Regulation Number 26 address foam manufacturing, a coil coating operation, and process heaters. NSPS J, NSPS Ja, NSPS T, MACT CC, MACT SSSS, MACT ZZZZ, and MACT DDDDD may also apply to and the above listed equipment and operations. However, the revisions to Regulation Number 26 apply on a broader basis.
- (II) The federal rules discussed in (I) are primarily technology-based in that they largely prescribe the use of specific technologies or work practices to comply.

- (III) The CAA establishes the 2008 and 2015 ozone NAAQS and requires Colorado to develop SIP revisions that will ensure attainment of the NAAQS. The ozone NAAQS was not determined taking into account concerns unique to Colorado. Similarly, EPA develops NSPS or NESHAP considering national information and data, not Colorado specific issues or concerns. In addition, Colorado cannot rely exclusively on a federally enforceable permit or federally enforceable NSPS or NESHAP to satisfy Colorado's ozone nonattainment area RACT obligations. Instead, Colorado can adopt applicable provisions into its SIP directly, as the Commission has done here.
- (IV) In addition to the 2008 NAAQS, Colorado must also comply with the lower 2015 ozone NAAQS. These current revisions may improve the ability of the regulated community to comply with new requirements needed to attain the lower NAAQS insofar as RACT analyses and efforts conducted to support the revisions adopted by the Commission may prevent or reduce the need to conduct additional RACT analyses for the more stringent NAAQS.
- (V) EPA has established Colorado's SIP RACT implementation deadlines. There is no timing issue that might justify changing the time frame for implementation of federal requirements.
- (VI) The revisions to Regulation Number 26 strengthen Colorado's SIP. These sections currently address emissions from foam manufacturing, a coil coating operation, and process heaters, while allowing for continued growth of Colorado's industry.
- (VII) The revisions to Regulation Number 26 establish reasonable equity for owners and operators subject to these rules by providing the same standards for similarly situated and sized sources.
- (VIII) If EPA does not approve Colorado's SIP, EPA may promulgate a Federal Implementation Plan; thus potentially determining RACT for Colorado's sources. This outcome may subject others to increased costs.
- (IX) Where necessary, the revisions to Regulation Number 26 include minimal monitoring, recordkeeping, and reporting requirements that correlate, where possible, to similar federal or state requirements.
- (X) Demonstrated technology is available to comply with the revisions to Regulation Number 26. Some of the revisions expand upon requirements already applicable. The revisions concerning major sources of NOx generally reflect current emission controls and work practices.

- (XI) As set forth in the Economic Impact Analysis, the revisions to Regulation Number 26 will reduce emissions in a cost-effective manner.
- (XII) Alternative rules could also provide reductions in ozone, VOC, and NO_x to help to attain the NAAQS. However, a no action alternative would very likely result in an unapprovable SIP.

Findings of Fact

To the extent that § 25-7-110.8, C.R.S., requirements apply to this rulemaking, and after considering all the information in the record, the Commission hereby makes the determination that

- (I) These rules are based upon reasonably available, validated, reviewed, and sound scientific methodologies, and the Commission has considered all information submitted by interested parties.
- (II) Evidence in the record supports the finding that the rules shall result in a demonstrable reduction of greenhouse gas and VOC emissions.
- (III) Evidence in the record supports the finding that the rules shall bring about reductions in risks to human health and the environment that justify the costs to implement and comply with the rules.
- (IV) The rules are the most cost-effective alternative to achieve the necessary reduction in air pollution and provide the regulated entity flexibility.
- (V) The selected regulatory alternative will maximize the air quality benefits of regulation in the most cost-effective manner.

IV. November 19-21, 2025 (Revisions to Part B, Sections II., VII., and VIII.)

This Statement of Basis, Specific Statutory Authority, and Purpose complies with the requirements of the State Administrative Procedure Act, § 24-4-101, C.R.S., et seq., the Colorado Air Pollution Prevention and Control Act (State Air Act or the Act), § 25-7-101, C.R.S., et seq., and the Air Quality Control Commission's (Commission) Procedural Rules, 5 C.C.R. §1001-1.

Basis

On July 24, 2024, EPA reclassified the Denver Metro/North Front Range (DM/NFR) and northern Weld County ozone nonattainment area to serious for the 2015 8-hour Ozone National Ambient Air Quality Standard of 70 parts per billion (ppb) (2015 ozone NAAQS). See 89 Fed. Reg. 59832.

To ensure progress towards attainment of the ozone NAAQS and respond to Colorado's requirements under the Clean Air Act (CAA), the Commission adopted revisions to include reasonably available control technology (RACT) for major sources of volatile organic compounds (VOC) and nitrogen oxides (NOx) in the nonattainment areas, specifically expanding requirements for combustion equipment at major NOx sources and major source industrial waste facilities in northern Weld County.

Specific Statutory Authority

The State Air Act, specifically § 25-7-105(1), C.R.S., directs the Commission to promulgate such rules and regulations as are consistent with the legislative declaration set forth in § 25-7-102, C.R.S., and that are necessary for the proper implementation and administration of Article 7. The Act broadly defines air pollutant to include essentially any gas emitted into the atmosphere (and, as such, includes VOC, NOx, methane and other hydrocarbons) and provides the Commission broad authority to regulate air pollutants. Section 105(1)(a)(I) directs the Commission to adopt a state implementation plan (SIP) to attain the NAAQS. Section 25-7-106, C.R.S., provides the Commission maximum flexibility in developing an effective air quality program and promulgating such combination of regulations as may be necessary or desirable to carry out that program. Section 25-7-106, C.R.S., also authorizes the Commission to promulgate emission control regulations applicable to the entire state, specified areas or zones, or a specified class of pollution. Section 25-7-106(6), C.R.S., further authorizes the Commission to require owners and operators of any air pollution source to monitor, record, and report information. Sections 25-7-109(1)(a) and (2) of the Act authorize the Commission to promulgate regulations requiring effective and practical air pollution controls for significant sources and categories of sources and emission control regulations pertaining to NOx and hydrocarbons.

Purpose

The following section sets forth the Commission's purpose in adopting the revisions to Regulation Number 26 and includes the technological and scientific rationale for the adoption of the revisions.

Major Source RACT

Due to the reclassification to serious for the 2015 ozone NAAQS, Colorado must submit revisions to its SIP to address the ozone nonattainment area requirements, as set forth in CAA §§ 172, 182(c), and the final SIP Requirements Rules (83 Fed. Reg. 62998 (Dec. 6, 2018)). Serious SIPs must include provisions that require the implementation of RACT for major sources of VOC and/or NO_x (i.e., sources that emit or have the potential to emit 50 tpy or more) and for each category of VOC sources covered by a Control Technique Guideline (CTG) for which Colorado has sources in the nonattainment area.

Therefore, to address the serious nonattainment area requirements under CAA § 182(c), the Commission adopted revisions to Regulation Number 26 to include RACT requirements in Colorado's ozone SIP for 50 tpy major sources of VOC and/or NO_x including expanding the engine, combustion equipment, and industrial waste requirements.

Specifically concerning engine and combustion equipment requirements, the Commission expanded the applicability of the control technology requirements for natural gas-fired stationary reciprocating internal combustion lean-burn engines in Part A, Section I.D.4.b. to engines located in northern Weld County for SIP purposes in Part A, Section I.A. These requirements already apply on a state-only basis. Similarly, the Commission expanded the applicability of the combustion equipment requirements in Part A, Section II.A. to major stationary sources of NO_x (50 tpy NO_x) located in northern Weld County. This expansion may impact boilers, turbines, compression ignition reciprocating internal combustion engines, and process heaters at subject sources. Lastly, the Commission expanded the applicability of the industrial waste facility requirements in Part A, Section VIII.A. to major stationary industrial waste facilities (50 tpy VOC) located in northern Weld County. This expansion will impact one facility and require work practices, recordkeeping, and reporting; aligning with current permit requirements.

The Commission also repealed requirements for major stationary poultry waste processing facilities. These provisions were adopted in 2022 for major sources (25 tpy) located in the 8-hour ozone control area. The subject source is no longer a major stationary source and, therefore, these SIP-RACT provisions are no longer required.

The Commission also made typographical, grammatical, and formatting corrections throughout the regulation.

Incorporation by Reference

The Commission will update regulatory references as needed as opportunities arrive.

Additional Considerations

Colorado must revise Colorado's ozone SIP to address the serious ozone nonattainment area requirements. The CAA does not expressly address all of the provisions adopted by the Commission. Rather, federal law establishes the ozone NAAQS and requires Colorado to develop a SIP adequate to attain the NAAQS. Therefore, the Commission adopted certain revisions to Regulation Number 26 to satisfy Colorado's nonattainment area obligations and further achieve reductions of ozone precursor emissions. These revisions do not exceed or differ from the federal act due to state flexibility in determining what control strategies to implement to reduce emissions. However, where the proposal may differ from federal rules under the federal act, in accordance with § 25-7-110.5(5)(b), C.R.S., the Commission determines:

- (I) The revisions to Regulation Number 26 address combustion equipment and industrial waste facilities. New Source Performance Standards (NSPS) GG, NSPS KKKK, NSPS IIII, NSPS JJJJ, National Emissions Standards for Hazardous Air Pollutants (NESHAP) HH, and NESHAP ZZZZ may also apply to and the above listed equipment and operations. However, the revisions to Regulation Number 26 apply on a broader basis.
- (II) The federal rules discussed in (I) are primarily technology-based in that they largely prescribe the use of specific technologies or work practices to comply.
- (III) The CAA establishes the ozone NAAQS and requires Colorado to develop SIP revisions that will ensure attainment of the NAAQS. The ozone NAAQS were not determined taking into account concerns unique to Colorado. Similarly, EPA develops NSPS or NESHAP considering national information and data, not Colorado specific issues or concerns. In addition, Colorado cannot rely exclusively on a federally enforceable permit or federally enforceable NSPS or NESHAP to satisfy Colorado's ozone nonattainment area RACT obligations. Instead, Colorado can adopt applicable provisions into its SIP directly, as the Commission has done here.
- (IV) Colorado must comply with the lower 2015 ozone NAAQS. These current revisions may improve the ability of the regulated community to comply with new requirements needed to attain the lower NAAQS insofar as RACT analyses and efforts conducted to support the revisions adopted by the Commission may prevent or reduce the need to conduct additional RACT analyses for the more stringent NAAQS.
- (V) EPA has established Colorado's SIP RACT implementation deadlines. There is no timing issue that might justify changing the time frame for implementation of federal requirements.

- (VI) The revisions to Regulation Number 26 strengthen Colorado's SIP. These sections currently address emissions from combustion equipment and industrial waste facilities, while allowing for continued growth of Colorado's industry.
- (VII) The revisions to Regulation Number 26 establish reasonable equity for owners and operators subject to these rules by providing the same standards for similarly situated and sized sources.
- (VIII) If EPA does not approve Colorado's SIP, EPA may promulgate a Federal Implementation Plan; thus, potentially determining RACT for Colorado's sources. This outcome may subject others to increased costs.
- (IX) Where necessary, the revisions to Regulation Number 26 include minimal monitoring, recordkeeping, and reporting requirements that correlate, where possible, to similar federal or state requirements.
- (X) Demonstrated technology is available to comply with the revisions to Regulation Number 26. Some of the revisions expand upon requirements already applicable. The revisions concerning major sources of NO_x generally reflect current emission controls and work practices.
- (XI) As set forth in the Economic Impact Analysis, the revisions to Regulation Number 26 will reduce emissions in a cost-effective manner.
- (XII) Alternative rules could also provide reductions in ozone, VOC, and NO_x to help to attain the NAAQS. However, a no action alternative would very likely result in an unapprovable SIP.

Findings of Fact

To the extent that § 25-7-110.8, C.R.S., requirements apply to this rulemaking, and after considering all the information in the record, the Commission hereby makes the determination that

- (I) These rules are based upon reasonably available, validated, reviewed, and sound scientific methodologies, and the Commission has considered all information submitted by interested parties.
- (II) Evidence in the record supports the finding that the rules shall result in a demonstrable reduction of NO_x and VOC emissions.
- (III) Evidence in the record supports the finding that the rules shall bring about reductions in risks to human health and the environment that justify the costs to implement and comply with the rules.

- (IV) The rules are the most cost-effective alternative to achieve the necessary reduction in air pollution and provide the regulated entity flexibility.
- (V) The selected regulatory alternative will maximize the air quality benefits of regulation in the most cost-effective manner.

Editor's Notes

History

New rule eff. 06/14/2023.

Part A rule I.C, Part B rules I.C.1, I.D.4.c Table A, I.D.5.a.(i)(B), I.D.5.a.(ii)(B), I.D.5.b.(i), I.D.5.b.(ii) Table 2, I.D.6, II.A.4.f.(i), II.A.4.g.(i) Table 2, II.A.5.b.(ii) (E)-(F), II.A.7.h, II.A.8.b.(i), III.D, V.A.1, V.A.4.a, Part C rule II eff. 02/14/2024.

Part B outline, rules I.D.4.c. Table B, I.D.6.a.(vi)(E), II.A.4, II.A.4.g.(i), II.A.4.g.(i) Table 2, II.A.4.g.(iv), II.A.4.g.(iv) Table 3, II.A.4.g.(v), II.A.4.g.(v) Table 4, II.A.5.a.(iii), II.A.5.b.(i)(A), II.A.5.b.(ii)(B)(1), II.A.8.c, V.A.8.b, IX, Part C III eff. 02/14/2025.
Part B rule III.D repealed eff. 02/14/2025.

Part B I.A.3-4, II.A.1.f, II.A.2, II.A.2.e, II.A.4, II.A.4.b(i), II.A.4.b(ii), II.A.4.g(iv), II.A.5.a(ii), II.A.5.a(iv), II.A.5.a(vi), II.A.5.b(iv), II.A.5.b(vi), II.A.6.b(viii)(D), II.A.6.b(viii)(E)-(H), II.A.6.c(ii), II.A.7.f(iii), II.A.8.a, II.A.8.b, II.A.8.d, VII, VIII.A.1, VIII.A.2.b-c, VIII.A.5, PART C IV eff. 01/14/2026.